STRENGTHENING HEALTH SYSTEMS AND RMNCAH-N OUTCOMES THROUGH RAPID CYCLE ANALYTICS AND DATA USE

Country Workshop: Introduction to FASTR RMNCAH-N Service Use Monitoring

Perth, Australia

Jan 10-12

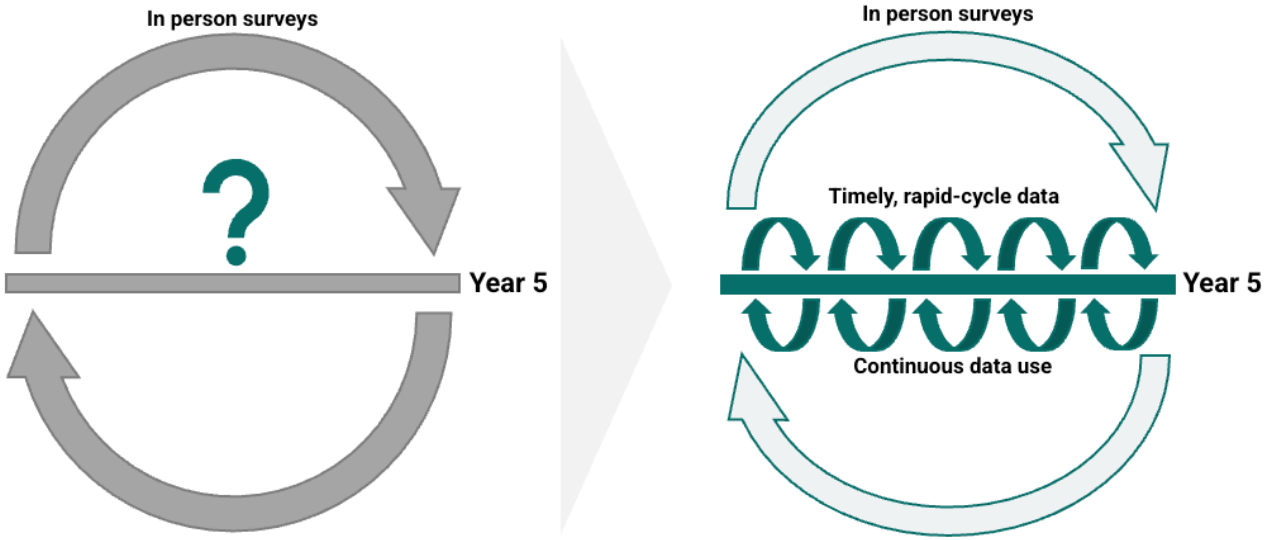
FREQUENT ASSESSMENTS & SYSTEMS TOOLS FOR RESILIENCE

Welcome and Opening Remarks

Introductions

What are we trying to achieve?

Rapid cycle analytics accelerates improvements in RMNCAH-N outcomes by increasing the systematic use of data for decision making



How can this be achieved?

Timely, rigorous, and low-cost approaches to monitoring PHC systems, underpinned by capacity building and data use support aligned with country demand and needs

RMNCAH-N Service Use

Leverage HMIS data to routinely monitor changes in the coverage of health services during reforms, shocks, or within vulnerable populations, while identifying and addressing data quality challenges.

Rapid-Cycle Health Facility Phone Surveys

Assess gaps in primary health care service delivery; understand the impact of health reforms or shocks on PHC performance.

Rapid-Cycle Household and Client Surveys

Generate snapshots of health service utilization and foregone care, and community perceptions of service quality

Digging deeper with follow-on analyses

Efficiently diagnosis underlying challenges and assess implementation contexts.

Build capacity + data use support

Building within-country capacity for data collection, analysis, and use including data reviews and subnational feedback loops.

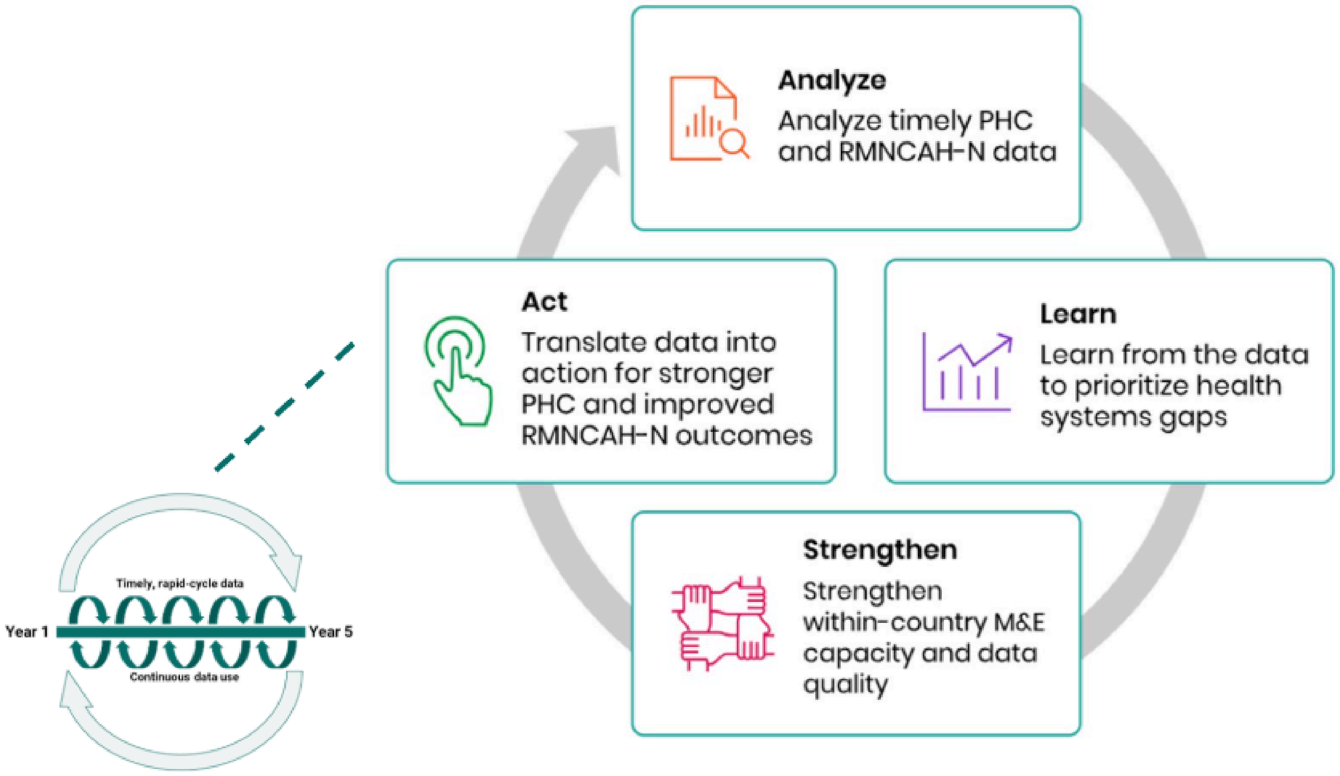
RMNCAH-N service use monitoring

Rapid-cycle approaches using routine HMIS data can:

- **Evaluate HMIS data quality** at national and sub-national levels
- **Measure monthly changes** in health service utilization
- **Compare coverage trends** with country targets

What is FASTR?

An approach to catalyzing continuous 'analyze, learn, strengthen, act' cycles to drive the systematic use of timely data for decision making.

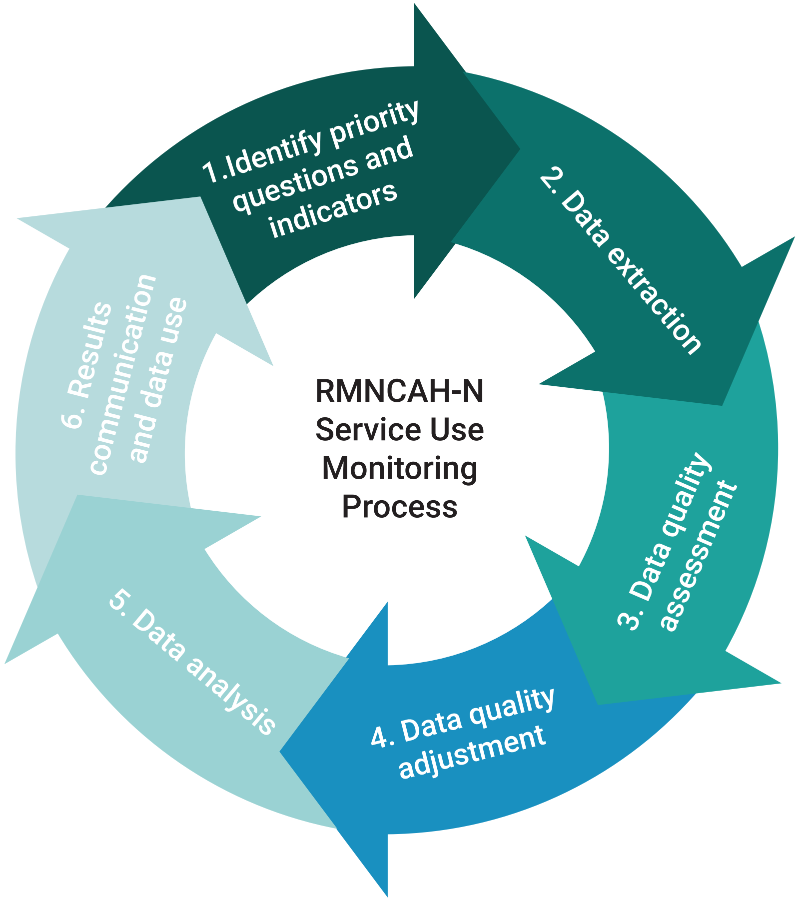


What is the FASTR approach to RMNCAH-N service use monitoring?

Quarterly analyses of DHIS2 data, focusing on prioritized national indicators

Building sustainable tools to ensure that stakeholders who need to use data can generate the right analysis and visualizations, at the right time, on their indicators of interest

Combining analysis and visualization with capacity strengthening and data use support for sustainability and institutionalization



Agenda

Day 1

Time	Agenda	Facilitator/Presenter
09:00	Title Slide	
	Workshop Objectives	
	Introduction to FASTR	
	Agenda	
	Data Extraction	
	Lunch Break	
	Identify Questions & Indicators	
	Tea Break	
	End of Day 1	
09:00	Day 2 Recap	
	FASTR Analytics Platform	

Agenda

Day 2

Time	Agenda	Facilitator/Presenter
	Data Quality Assessment	
	Tea Break	
	Data Quality Adjustment	
	Tea Break	
	End of Day 2	
	Day 3 Recap	
	Data Analysis	
	Lunch Break	
	Results Communication	
	Tea Break	
	Closing	
	Data Quality Assessment	

Show of hands...

Do you regularly extract data from DHIS2?

If so, what are the primary reasons?

Why would you extract data from DHIS2? Why not just do analysis in DHIS2 itself?

Data quality adjustment

The FASTR approach focuses on data quality adjustments to expand the analyses countries can do with DHIS2 data and to generate more robust estimates.

Analysis complexity

The FASTR approach uses more advanced statistical methods, such as regression analysis, which are not available in DHIS2. While DHIS2 can plot trends over time using raw data, FASTR can go further by identifying significant increases or decreases in service volume, adjusting for data quality issues, accounting for expected seasonal variations, and comparing key periods, such as before and after a reform.

The choice between DHIS2 and the FASTR approach should be guided by the specific purpose of your analysis. Select the tool that best aligns with your analytical needs!

Data format and granularity

Data format example

Data should be downloaded for each indicator of interest, at facility level, and monthly for the period of interest.

Data should be saved in long format meaning each row represents a single observation or measurement (see example).

Data should be saved in .csv format and can be saved in either a single .csv file or multiple .csv files which will be combined when uploading to the analysis platform.

How much data?

Initial FASTR analysis

- Generally recommended to download approximately five years of historical data
- However, the exact period should be determined based on data availability, consistency in indicator definitions over time, and the specifics of a country's routine data system
- Ideally, using at least five years of historical data allows for a thorough assessment of trends over time

Routine update to FASTR analysis

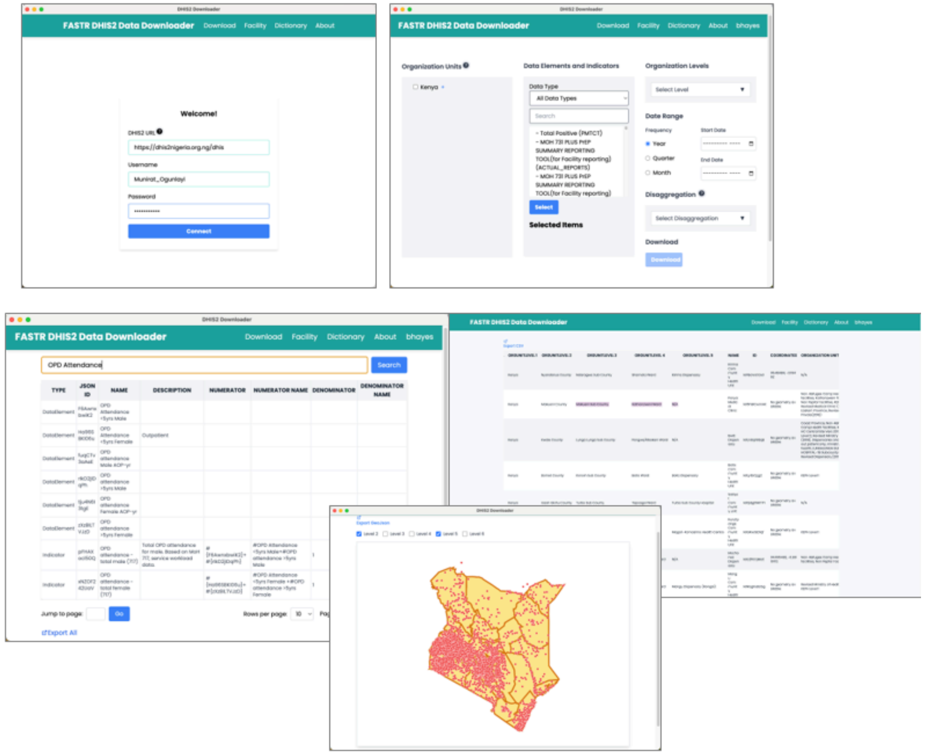
- Start with the existing database and download new data covering the most recent months not previously included – this is usually a three-month period when the FASTR analysis is being implemented on a quarterly basis
- Additionally, include the three proceeding months to the new data time period, as this relatively recent data is often subject to changes due to late reporting or data quality adjustments
- If you have reason to believe there have been substantial changes to the historical data, you can always choose to redownload a longer time period

Data extraction

We offer two tools for bulk DHIS2 data extraction: a user-friendly Data Downloader and a direct import feature within the FASTR analytics platform.

The Data Downloader provides a streamlined interface to download DHIS2 data. This tool is particularly useful to explore DHIS2 metadata and download indicators requiring disaggregated dimensions.

The Data Downloader is available at: https://github.com/worldbank/DHIS2-Downloader/releases/



DHIS2 Data Downloader

The Data Downloader is a desktop application for extracting data from DHIS2.

Key features:

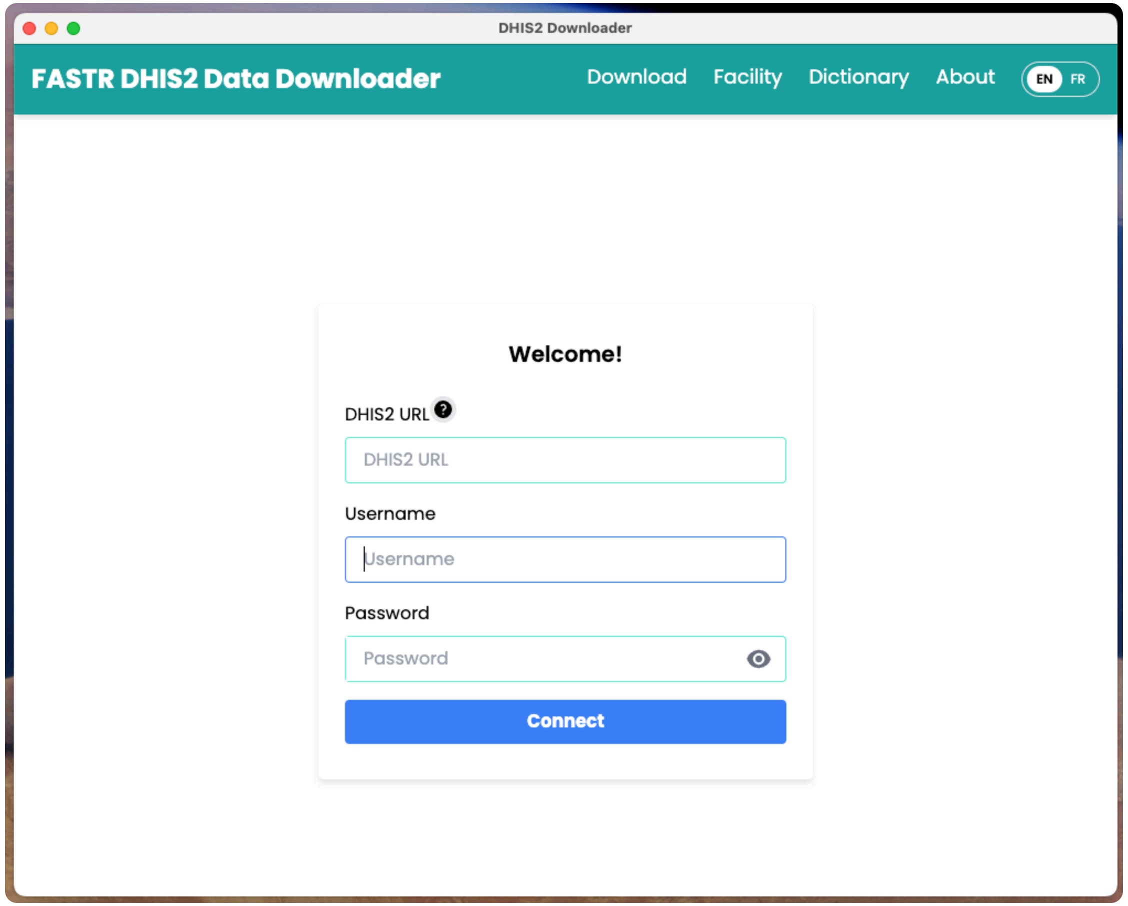
- Connect to any DHIS2 instance
- Browse and select data elements and indicators
- Download facility-level data in CSV format
- Maintain download history

Download from GitHub:

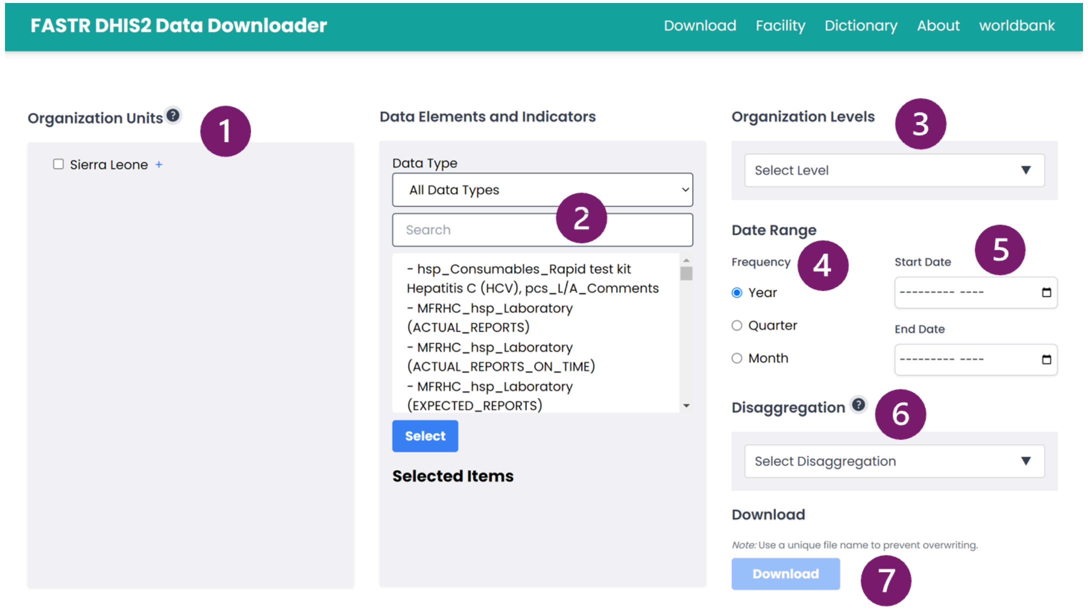
https://github.com/worldbank/DHIS2-Downloader/releases/

Facilitator will demonstrate the Data Downloader

Data Downloader: Login



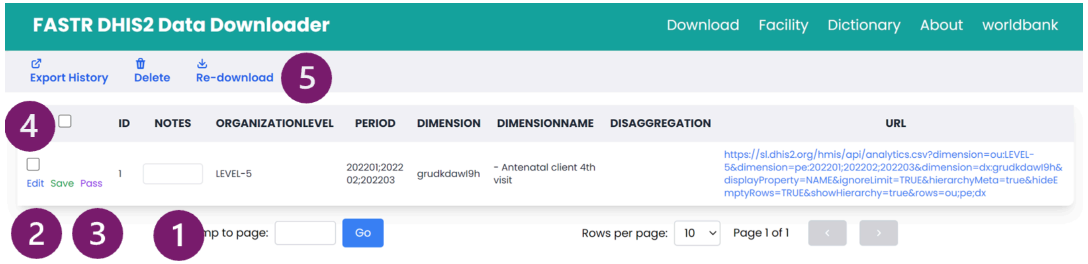
Data Downloader: Overview



Main interface

- Browse available data elements and indicators
- Select time periods and organization units
- Configure download options
- Start data extraction

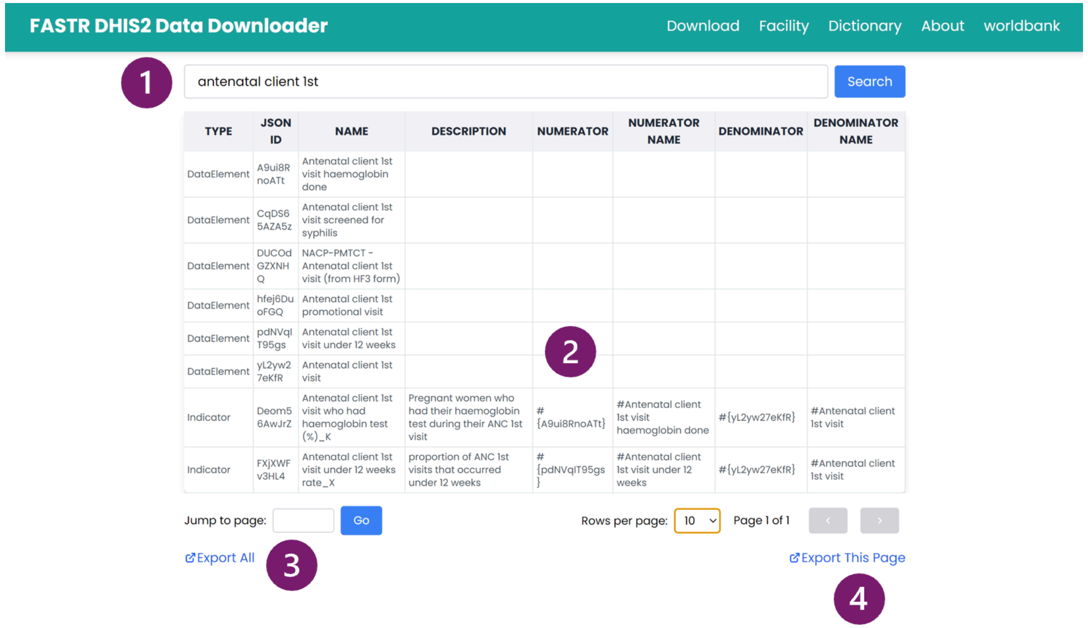
Data Downloader: Download history



Track your downloads

- View all previous download sessions
- Re-download data with same parameters
- Access download logs and status
- Manage downloaded files

Data Downloader: Data dictionary

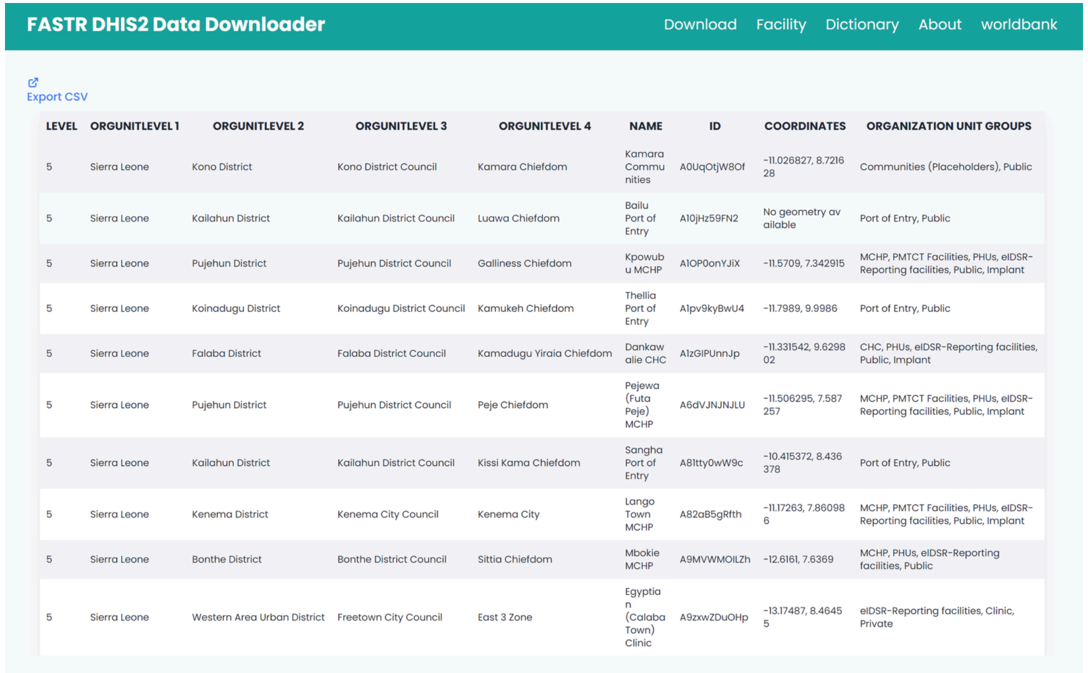


TYPE	JSON ID	NAME	DESCRIPTION	NUMERATOR	NUMERATOR NAME	DENOMINATOR	DENOMINATOR NAME
DataElement	A9ui8RnoATt	Antenatal client 1st visit haemoglobin done					
DataElement	CqDS65AZA5z	Antenatal client 1st visit screened for syphilis					
DataElement	DUCOdGZXNHQ	NACP-PMTCT - Antenatal client 1st visit (from HF3 form)					
DataElement	hfej6DuoFGQ	Antenatal client 1st promotional visit					
DataElement	pdNVqlT95gs	Antenatal client 1st visit under 12 weeks					
DataElement	yL2yw27eKfR	Antenatal client 1st visit					
Indicator	Deom56AwJrZ	Antenatal client 1st visit who had haemoglobin test (%)_K	Pregnant women who had their haemoglobin test during their ANC 1st visit	#{A9ui8RnoATt}	#Antenatal client 1st visit haemoglobin done	#{yL2yw27eKfR}	#Antenatal client 1st visit
Indicator	FXjXWFv3HL4	Antenatal client 1st visit under 12 weeks rate_X	proportion of ANC 1st visits that occurred under 12 weeks	#{pdNVqlT95gs}	#Antenatal client 1st visit under 12 weeks	#{yL2yw27eKfR}	#Antenatal client 1st visit

Explore available data

- Browse all data elements from your DHIS2
- Search by name or code
- View metadata and definitions
- Identify indicators for your analysis

Data Downloader: Facility list

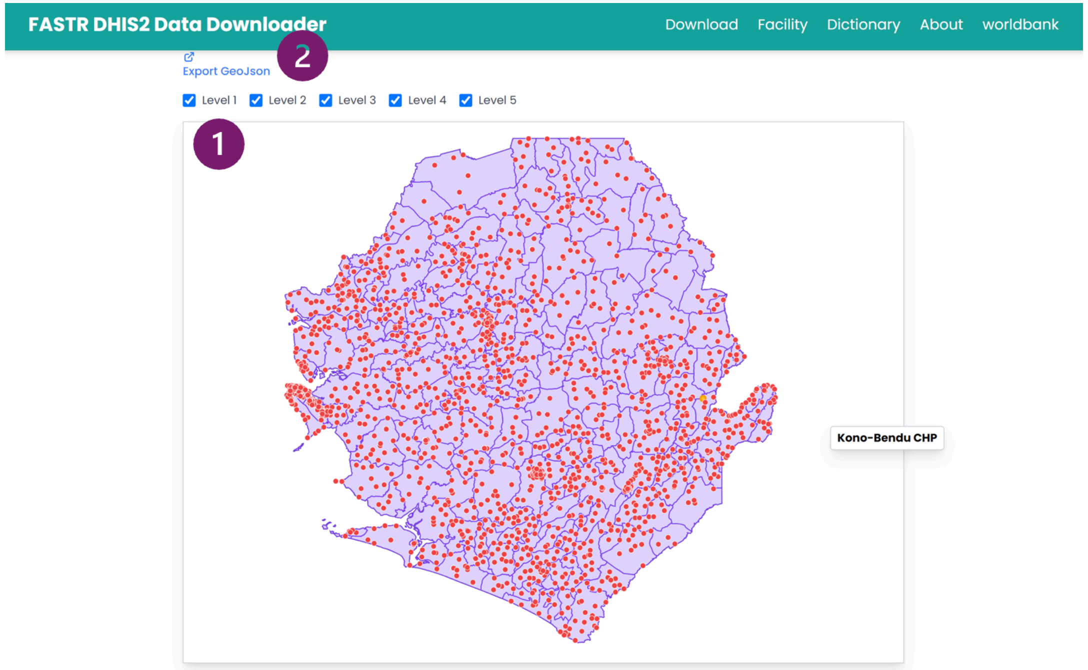


LEVEL	ORGUNITLEVEL 1	ORGUNITLEVEL 2	ORGUNITLEVEL 3	ORGUNITLEVEL 4	NAME	ID	COORDINATES	ORGANIZATION UNIT GROUPS
5	Sierra Leone	Kono District	Kono District Council	Kamara Chiefdom	Kamara Communities	A0UqOtjW8Of	-11.026827, 8.721628	Communities (Placeholders), Public
5	Sierra Leone	Kailahun District	Kailahun District Council	Luawa Chiefdom	Bailu Port of Entry	A10jHz59FN2	No geometry available	Port of Entry, Public
5	Sierra Leone	Pujehun District	Pujehun District Council	Galliness Chiefdom	Kpowubu MCHP	A1OP0onYJiX	-11.5709, 7.342915	MCHP, PMTCT Facilities, PHUs, eIDSR-Reporting facilities, Public, Implant
5	Sierra Leone	Koinadugu District	Koinadugu District Council	Kamukeh Chiefdom	Thellia Port of Entry	A1pv9kyBwU4	-11.7989, 9.9986	Port of Entry, Public
5	Sierra Leone	Falaba District	Falaba District Council	Kamadugu Yiraia Chiefdom	Dankawalie CHC	A1zGIPUnnJp	-11.331542, 9.629802	CHC, PHUs, eIDSR-Reporting facilities, Public, Implant
5	Sierra Leone	Pujehun District	Pujehun District Council	Peje Chiefdom	Pejewa (Futa Peje) MCHP	A6dVJNJNJLU	-11.506295, 7.587257	MCHP, PMTCT Facilities, PHUs, eIDSR-Reporting facilities, Public, Implant
5	Sierra Leone	Kailahun District	Kailahun District Council	Kissi Kama Chiefdom	Sangha Port of Entry	A81tty0wW9c	-10.415372, 8.436378	Port of Entry, Public
5	Sierra Leone	Kenema District	Kenema City Council	Kenema City	Lango Town MCHP	A82aB5gRfth	-11.17263, 7.860986	MCHP, PMTCT Facilities, PHUs, eIDSR-Reporting facilities, Public, Implant
5	Sierra Leone	Bonthe District	Bonthe District Council	Sittia Chiefdom	Mbokie MCHP	A9MVWMOILZh	-12.6161, 7.6369	MCHP, PHUs, eIDSR-Reporting facilities, Public
5	Sierra Leone	Western Area Urban District	Freetown City Council	East 3 Zone	Egyptian (Calaba Town) Clinic	A9zxwZDuOHp	-13.17487, 8.46455	eIDSR-Reporting facilities, Clinic, Private

Facility management

- View complete facility list
- Filter by administrative level
- Search by facility name
- Export facility data

Data Downloader: Facility map



Geographic visualization

- Download GeoJSON boundary files
- Toggle administrative boundaries by level (Level 1 = country, Level 2 = regions, etc.)
- Higher levels display facility points
- Useful for verifying geographic structure

🍽️ Lunch Break

60 minutes

Back at

What is a data use case?

A data use case is a specific scenario where data is utilized to achieve a particular goal or solve a problem.

Why is defining a data use case important?

- Guides decision making by providing a clear framework for analysis
- Enhances efficiency by focusing analyses on a set of relevant key indicators to solve a specific data need
- Leads to better results by aligning data efforts with organizational goals

What is our common data use case that will be the focus of this workshop?

Following large shifts in resource availability from external sources, many countries are experiencing abrupt and dramatic reductions in financing

- Resulting in critical gaps in programs and systems
- Leading to potentially severe effects on service delivery and health outcomes for women, children and adolescents

Key questions arising:

- What is the magnitude of the cuts, and what effect are they having on service delivery?
- What is the optimal way to prioritize remaining resources?
- What other adaptations can safeguard and strengthen essential service delivery for women, children and adolescents?

How do we select indicators? What makes a good indicator for the FASTR analysis?

Indicator selection is critical to the quality and usefulness of FASTR analysis. Indicators should be chosen based on the following criteria:

- **Relevance** - Does this indicator provide data that aligns with our priority questions and objectives?
- **Volume** - Is this indicator collected at a high volume, which improves the robustness of analysis?
- **Completeness** - Does the indicator have a high completeness rate across reporting facilities?
- **Frequency** - Is the indicator reported frequently enough (e.g., monthly) to support rapid-cycle analysis?
- **Type** - Is this indicator a count of services delivered?

Development of a data use case

Content to be developed

This section will cover:

- Co-creation workshop approach with MoH and stakeholders
- Data use case development guidance
- Example use cases from country implementations

Defining priority questions

Effective data use relies on well-defined questions. Priority questions will guide the FASTR analysis and enhance decision making support.

Qualities of a good question:

- **Addresses a priority issue**: A topic of interest to you and policy makers
- **Relevant**: Important enough to be worth answering
- **Related to experiences that are alive**: Connected to current issues
- **Important to individuals/groups**: Matters to stakeholders
- **Answerable**: Can be addressed with available data and methods

Is my question a relevant priority? 5+ Ws to consider

- **Who** is your audience?
- **What** do they need and want to know?
- **When** do they need to know it by?
- **When** is the event/intervention/period they are interested in?
- **Why** do they need to know?
- **How** will they use the findings?

What do we mean by answerable?

We have the data

- Type, quantity, quality sufficient for the question

We have the analysis tools/methods

- Statistically valid; feasible to use

We have the time

- We can answer the question on a quarterly basis

PICO framework for identifying answerable questions

A standard tool from evidence-based medicine and public health research for formulating clear, answerable questions.

Component	Description
Population	Who is being investigated
Intervention	What is being investigated
Comparison	What is baseline/non-intervention
Outcome	What is public health objective

What makes a good indicator for FASTR analysis?

- **Relevance**: Does this indicator align with our priority questions and objectives?
- **Volume**: Is this indicator collected at a high volume, improving robustness of analysis?
- **Completeness**: Does the indicator have a high completeness rate across reporting facilities?
- **Frequency**: Is the indicator reported frequently enough (e.g., monthly) to support rapid-cycle analysis?
- **Type**: Is this indicator a count of services delivered?

Why focus on high-volume indicators?

One of the core strengths of the FASTR approach is its ability to adjust for data quality issues. High-volume indicators are better suited to this process because:

Reduced sensitivity to outliers
In low-volume indicators, individual data points can disproportionately affect trends.

More stable estimates
High-volume data reduce random variability and improve the reliability of trend detection.

Clearer identification of true anomalies
Larger counts make it easier to distinguish genuine outliers from natural variation.

Why focus on high-completeness indicators?

Indicators with high reporting completeness are preferred because they:

Improve data reliability
More complete data reduces bias and provide a more representative picture of service delivery.

Support consistent analysis
High completeness enables meaningful comparisons across time and geographic areas.

Reduce misinterpretation
Incomplete data can falsely suggest changes in service utilization when changes are driven by reporting gaps rather than real trends.

While statistical methods such as imputation can be used to address incomplete data, these methods require assumptions about missing values.

Why focus on count indicators?

Limitations of proportion indicators

- Proportions limit the ability to adjust numerators and denominators separately for data quality issues
- Numerators and denominators may each be affected by different sources of error
- Separating counts from denominator estimation allows for more transparent and flexible adjustment

Mortality as a rare event

- Mortality indicators are typically low-frequency and not well suited to adjustment
- These indicators are generally better analyzed using annual rather than monthly or quarterly data

FASTR core indicators

The FASTR approach focuses on a core set of RMNCAH-N indicators that represent key points along the reproductive, maternal, newborn, child, and adolescent health and nutrition continuum in low- and middle-income countries.

These indicators typically have higher reporting volumes and completeness and serve as proxies for broader service delivery patterns.

- Antenatal client 1st visit
- Antenatal client 4th visit
- Institutional delivery
- Postnatal care 1
- BCG doses
- Pentavalent 1st dose
- Pentavalent 3rd dose
- Outpatient visits

Countries have selected indicators to align with the disruptions context and country priorities

FASTR	Indicator mapping for RMNCAH-N Service Use Monitoring using Administrative Data					
Data Source Mapping				Indicator Definition		
INDICATOR CATEGORY	INDICATOR OF INTEREST	INDICATOR SOURCE (national strategy, WB project, FASTR core indicators, etc)	IS THIS INDICATOR IN DHIS2?	OFFICIAL INDICATOR NAME IN DHIS2	INDICATOR JSON ID	NOTES
RMNCAH+N	ANC1	FASTR core indicators				
RMNCAH+N	ANC4	FASTR core indicators				
RMNCAH+N	Deliveries - institutional	FASTR core indicators				
RMNCAH+N	BCG	FASTR core indicators				
RMNCAH+N	PNC1	FASTR core indicators				
RMNCAH+N	Penta1	FASTR core indicators				
RMNCAH+N	Penta3	FASTR core indicators				
RMNCAH+N	OPD visits	FASTR core indicators				

The FASTR Data Prep Checklist has been shared with countries.

The checklist includes the FASTR core RMNCAH-N indicators.

Countries have added additional indicators that may be particularly relevant to the country context (e.g., indicators related to services expected to be impacted by recent funding shifts, priority indicators for the government and/or the WB project).

These are the indicators included in the current analysis. Countries can continue to add indicators over time as needed for their use case(s).

Preparing for data extraction

Content to be developed

This section will cover:

- Pre-extraction checklist
- Understanding your DHIS2 configuration
- Mapping indicators to data elements
- Planning your extraction timeline

☕ Tea Break

15 minutes

Back at

See You Tomorrow!

Day 1 Complete

We resume tomorrow at **09:00**

FASTR analytics platform

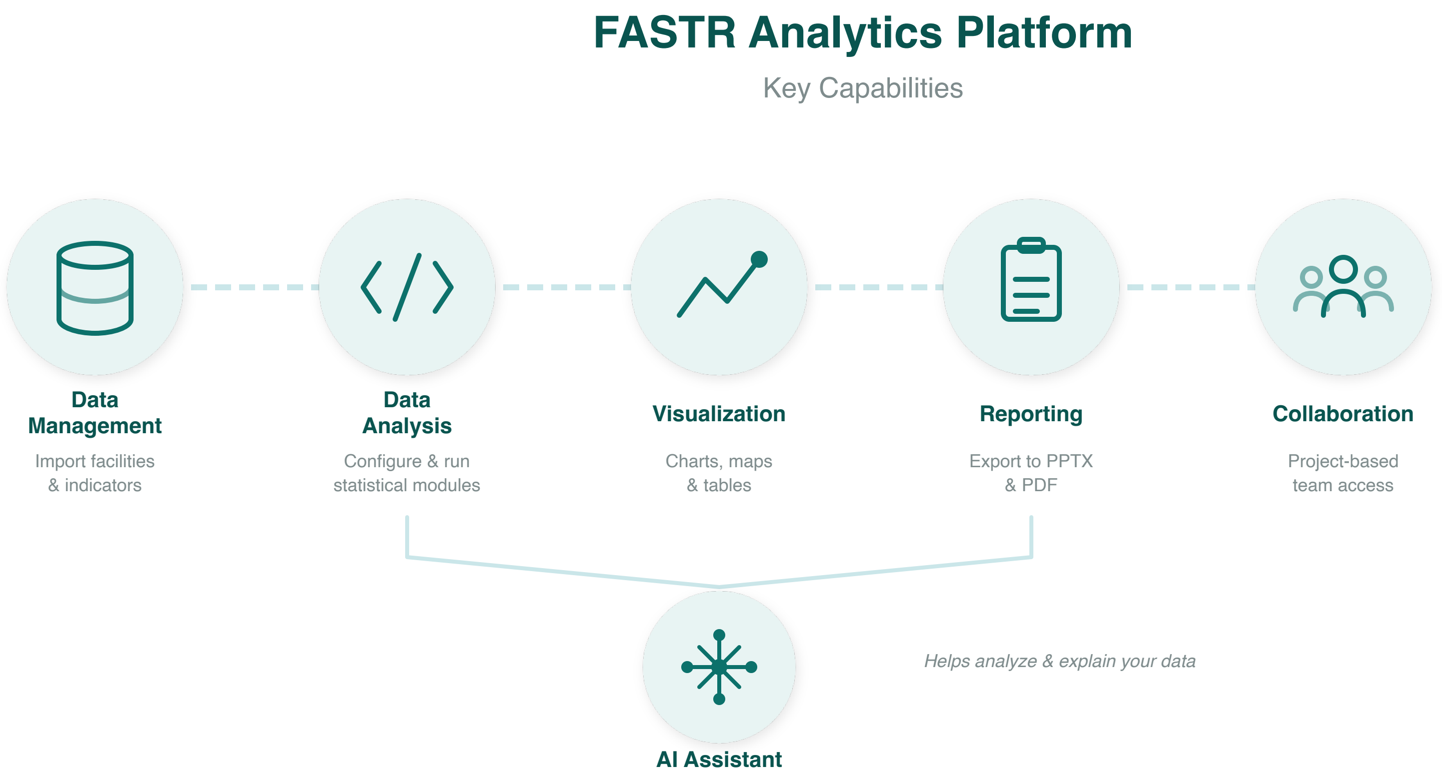
The FASTR analytics platform is a web-based tool designed to support data quality assessment, adjustment, and analysis for routine health data.

It allows users to upload and analyze data from various sources, including DHIS2, with built-in statistical methods to generate an adjusted dataset and run priority analyses on selected indicators.

The platform provides a user-friendly interface for running analyses and offers flexible options for visualizing and exporting results.

FASTR platform

Platform Capabilities



Data flows from import through analysis to shareable outputs.

Live Demo: Platform Access & Roles

In this demo, we will:

- Navigate to the FASTR platform
- Explore user roles: Administrator, Editor, Viewer
- Review user management and permissions
- Understand the workflow for uploading data and making analytical decisions

Facilitator will demonstrate in the live platform

Country Instance

Each country has its own **instance** of the FASTR analytics platform.

An instance contains:

- All registered users and their accounts
- The shared administrative structure (regions, districts, facilities)
- Indicator definitions and data sources
- All projects created for that country

Think of an instance as your country's dedicated workspace.

User Roles and Permissions

There are two levels of permissions in the platform:

Instance-level roles:

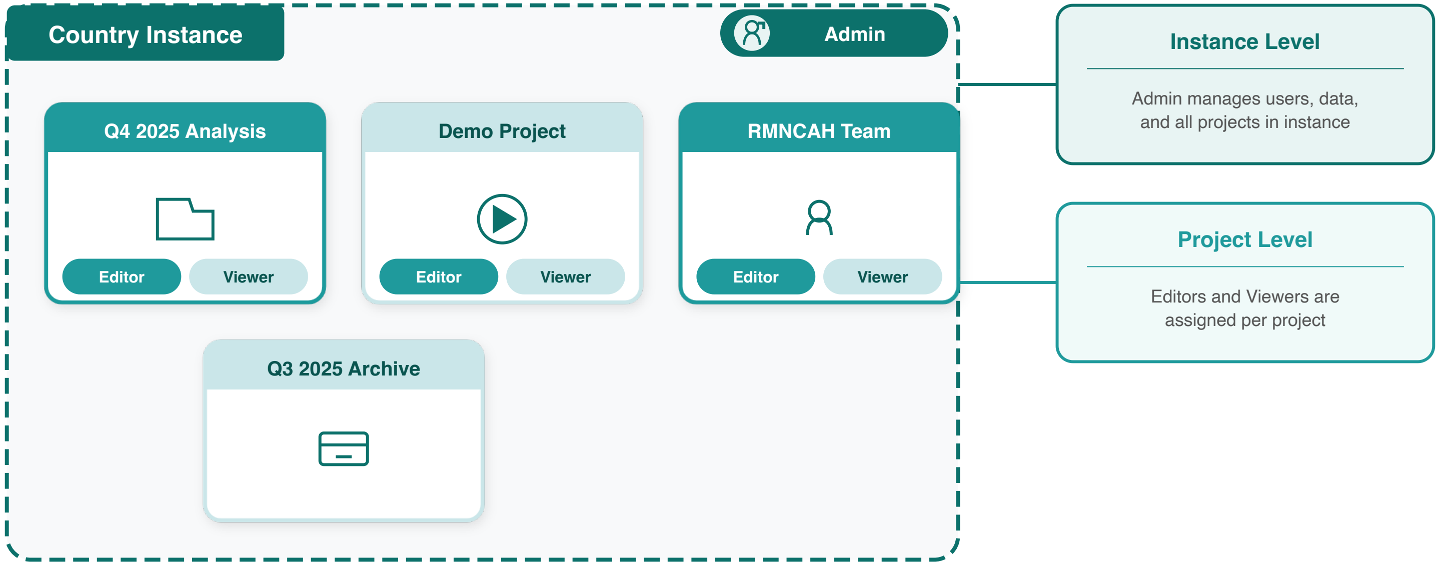
- **Instance Administrators** can add users, create projects, assign roles, upload data, import and configure modules, and run analyses

Project-level roles:

- **Project Editors** can create visualizations, create reports, and download/export results
- **Project Viewers** can view visualizations, view reports, and download/export results

Administrators are assigned per instance; Editors and Viewers are assigned per project.

Projects Within an Instance



Each country instance can contain **multiple projects**.

A country may only need one project, or multiple projects can be used for:

- Different versions of analyses
- A demo or playground project
- Separate projects for different teams or programs

Key questions when setting up:

- Who is the admin?
- Who can edit?
- Who can view?

Practice: Logging Into the Platform

Login page	Sign up form
1. Go to https://australia.fastr-analytics.org	**2.** Click Sign up and enter your details
3. Enter your information (verify email)	**4.** After login, you'll be added to a project

Configuring the analysis platform

- Configuration of the analysis platform is an admin feature
- We will work together to configure the following items:
    - Admin areas (regions, districts)
    - Facility structure
    - Indicator definitions
- Note since this is an admin feature all participants will NOT be doing this step. Instead, you will select one person to have admin rights, and they will help us walk through these steps.

Activity: Setting Up Admin Areas

In this hands-on session, we will configure:

- Admin areas (regions, districts)
- Facility structure
- Indicator definitions

Participants will work directly in the platform

Activity: Importing Data

In this hands-on session, we will:

- Review data format requirements
- Walk through the import process
- Handle validation and error checking

Participants will import their country's data

Activity: Installing and Running Modules

In this hands-on session, we will:

- Review available analysis modules
- Install required modules
- Run initial analyses

Participants will configure and run modules on their data

Activity: Creating a Project

In this hands-on session, we will:

- Set up a new project
- Configure project settings
- Select indicators and time periods
- Apply best practices for project organization

Participants will create their first project

Activity: Creating Visualizations

In this hands-on session, we will:

- Explore available chart types
- Create and customize visualizations
- Export charts for use in reports

Participants will build visualizations from their analysis

Activity: Creating Reports

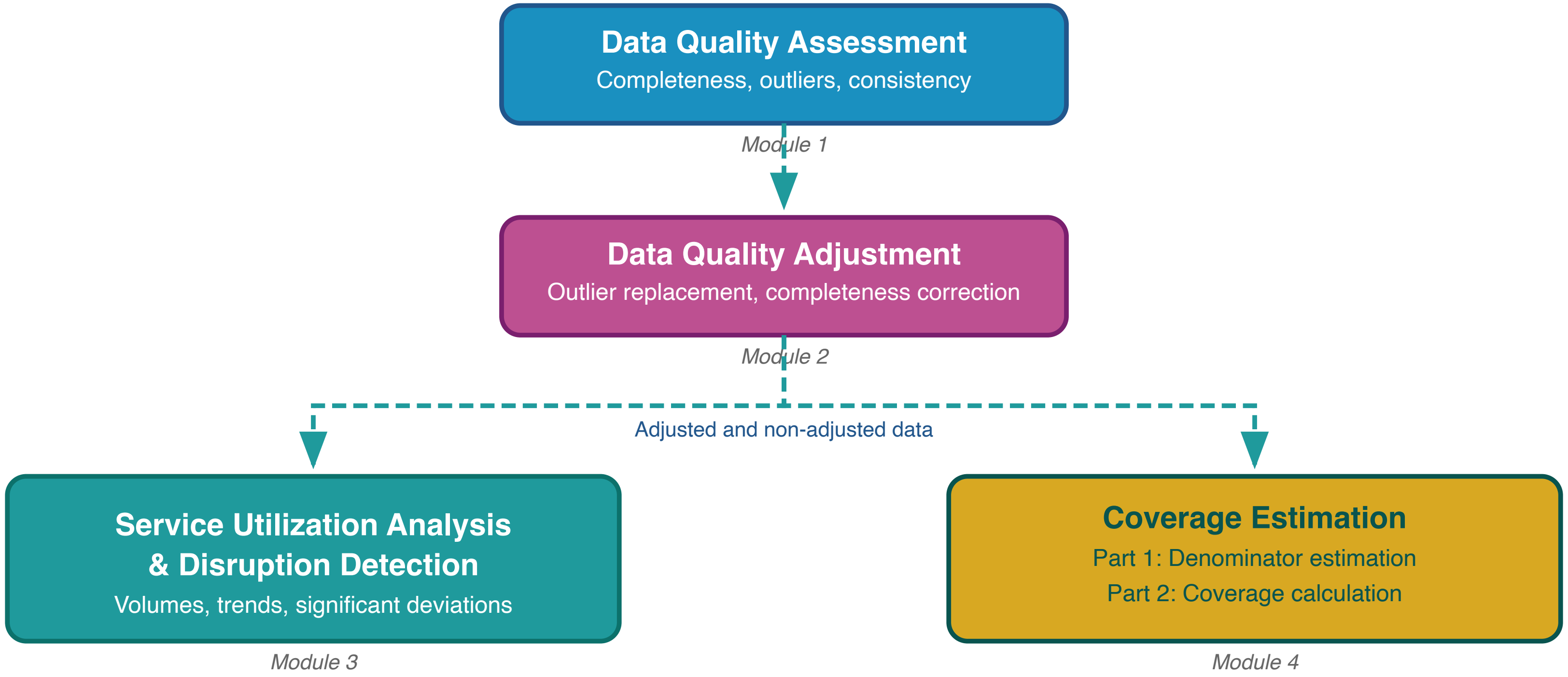
In this hands-on session, we will:

- Use report templates
- Generate automated reports
- Customize report content and layout

Participants will create their first quarterly report draft

Day 2

FASTR analytical pipeline



The FASTR analysis follows a sequential workflow where each step builds on the previous:

1. **Assess data quality** - Identify issues with completeness, outliers, and consistency
2. **Adjust for quality issues** - Apply corrections to improve data reliability
3. **Analyze adjusted data** - Generate service utilization and coverage estimates

FASTR takes a multi-pronged approach to data quality, with the belief that data quality should not be a barrier to data use – with the right feedback loops, use of data can contribute to improved quality

- We do granular data quality assessments and adjustments based on facility-level data leveraging HMIS access with an API
- We use only high-volume indicators, because the most-used services provide more stable estimates
- We focus on variations across time and space rather than specific point estimates and discuss the interpretation and relevance for decision-making with in-country decision-makers
- We believe that using the data and providing feedback is the first step to improving the data

We will discuss each of these areas over the next sessions.

Rationale for data quality assessment

Challenge: Routine health facility data may contain quality limitations:

- Reported values may fall outside plausible ranges
- Reporting gaps affect data completeness
- Inconsistencies exist between related indicators

Implications: Data quality limitations affect decision-making

- Inaccurate assessments of service delivery trends
- Misidentification of areas requiring intervention
- Suboptimal resource allocation

Assessing and adjusting for data quality

Quality assessments identify the highest priority issues and necessary analytical adjustments, so quality issues do not become a barrier to data analysis and use.

Objective 1: Enable analytical adjustment

Systematic data quality assessment supports the application of targeted adjustments, enhancing the utility of HMIS data for evidence-based decision-making.

Objective 2: Monitor data quality trends

Data quality assessment enables ongoing monitoring to:

- Inform indicator selection based on quality profiles across the HMIS
- Guide targeted data quality interventions and supportive supervision in areas with weaker data quality
- Evaluate the effectiveness of data quality improvement initiatives over time

Measures of data quality

Measures of data quality

Measures of data quality - detailed (1/2)

Domain	What does it measure?	How is it assessed?
Completeness	Are all data present?	Reporting completeness: whether all units report. Indicator completeness: whether values are recorded for specific data elements
Timeliness	Are data regularly submitted on time?	Whether units submitted reports before the set deadline

Measures of data quality - detailed (2/2)

Domain	What does it measure?	How is it assessed?
Consistency	Are data plausible in view of what has been previously reported?	Presence of outliers, consistency over time, consistency between related indicators, external comparison with other data sources, consistency of population data
Accuracy	Do data faithfully reflect actual service delivery?	Review of source documents and comparison to monthly reports and HMIS values (data verification factor)

How does FASTR data quality analysis differ from DHIS2? (1/2)

Purpose of data quality assessment

- **DHIS2:** focuses on data quality assessment to routinely strengthen data quality over time
- **FASTR:** focuses on assessing data quality to inform an analysis which answers a pressing policy question

Data quality adjustment

- **DHIS2:** focus is on identifying data quality issues and working with facilities to improve reporting practices
- **FASTR:** focus on applying analytical adjustment techniques to account for data quality issues in the analysis; goal is to generate the most robust estimates despite data quality challenges

How does FASTR data quality analysis differ from DHIS2? (2/2)

Selection of indicators, measures, and thresholds – FASTR focuses on DQA elements most relevant for analysis

- The purpose of the data quality assessment guides the selection of indicators, measures, and thresholds
- DHIS2 allows configuration of a DQA dashboard for any selection of indicators; FASTR selects indicators that will be used in a specific analysis
- DHIS2 DQA includes timeliness as a measure of data quality. FASTR does not include timeliness - it is important for strengthening routine reporting but less important for analysis with available data
- DHIS2 DQA includes reporting completeness and indicator completeness while FASTR focuses only on indicator completeness

Indicator completeness

What it measures: The extent to which facilities report data on selected core indicators

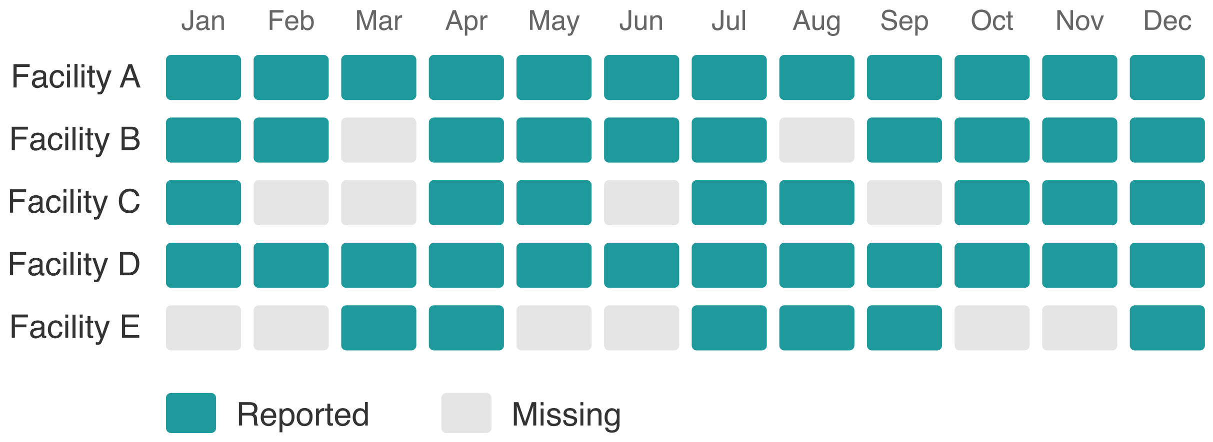
Why it matters:

- Higher completeness improves data reliability
- Stability over time strengthens trend analysis

Key distinction:
Indicator completeness ≠ reporting completeness. This metric examines specific data elements, not just whether the monthly form was submitted.

Region A: Indicator Completeness

5 health facilities reporting monthly on ANC1



5 facilities × 12 months = 60 expected
48 reported → 80% completeness

Definition of indicator completeness

For the FASTR analysis, completeness is defined as:

The percentage of reporting facilities each month out of the total number of facilities expected to report.

- A facility is deemed to be "reporting" if there is a non-missing, non-zero value recorded for the indicator and month
- A facility is expected to report if it has reported any volume for that indicator anytime within a year
- Facilities that do not report for six or more consecutive months at the beginning or end of their reporting period are classified as **inactive** rather than incomplete. This prevents penalizing facilities that have not yet begun reporting or have permanently ceased operations

Notes on completeness

- A high level of completeness does not necessarily indicate that the HMIS is representative of all service delivery in the country as some services may not be delivered in facilities, or some facilities may not report
- For countries where the DHIS2 system does not store 0's, indicator completeness may be underestimated if there are many low-volume facilities for a given indicator

Completeness: Percent of monthly values that are complete

For a given indicator in a given time period, the percent of monthly values that are complete:

% complete = # monthly values that are complete / total N of monthly values

Indicator Completeness

Percentage of facility-months with complete data, Jan 2022 to Apr 2025

	Antenatal care 1	Antenatal care 4	Institutional delivery	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG vaccine	Penta vaccine 1	Penta vaccine 3	Outpatient visit
District 001	93.9%	90.3%	83.3%	70.8%	82.8%	92.3%	91.9%	90.4%	94.9%
District 002	88.5%	85.9%	82.0%	64.2%	79.8%	90.8%	88.6%	88.1%	88.4%
District 003	89.8%	87.0%	85.4%	53.4%	78.5%	86.2%	86.3%	85.5%	88.3%
District 004	90.7%	82.6%	84.5%	67.6%	82.1%	89.1%	90.4%	89.7%	90.4%
District 005	89.7%	84.0%	81.3%	60.4%	80.9%	84.3%	84.2%	83.6%	89.5%
District 006	85.5%	79.4%	74.5%	56.8%	74.5%	85.3%	82.9%	79.2%	85.6%
District 007	97.2%	93.8%	91.8%	73.5%	90.8%	91.3%	95.2%	87.9%	97.0%
District 008	93.1%	87.6%	85.5%	44.6%	71.8%	79.9%	87.2%	87.5%	95.0%
District 009	95.7%	75.8%	85.0%	34.9%	77.6%	83.5%	92.2%	78.1%	95.9%
District 010	99.2%	99.6%	95.8%	98.4%	94.2%	94.6%	93.2%	88.7%	100.0%
District 011	98.0%	95.2%	94.4%	66.6%	91.7%	96.1%	95.4%	91.8%	98.2%
District 012	92.6%	86.4%	83.7%	64.5%	84.7%	84.7%	87.2%	86.7%	92.6%
District 013	93.4%	90.8%	90.7%	77.1%	88.2%	90.3%	90.8%	87.7%	93.1%
District 014	88.8%	79.1%	81.2%	77.9%	81.2%	90.0%	89.8%	86.3%	91.2%
District 015	93.0%	88.2%	85.3%	63.9%	83.1%	86.7%	88.8%	88.7%	93.1%
District 016	96.0%	85.1%	92.0%	65.7%	91.6%	96.1%	95.5%	88.6%	96.8%
District 017	90.1%	84.8%	90.4%	53.4%	83.0%	72.7%	79.3%	76.0%	89.5%
District 018	97.1%	95.9%	93.2%	79.8%	92.3%	95.4%	95.8%	92.9%	97.9%

90% or above
80% to 89%
Below 80%

Higher completeness improves the reliability of the data, especially when completeness is stable over time. Completeness is defined as the percentage of reporting facilities each month out of the total number of facilities expected to report. A facility is expected to report if it has reported any volume for each indicator anytime within a year. A high completeness does not indicate that the HMIS is representative of all service delivery in the country, as some services may not be delivered in facilities, or some facilities may not report.

Outliers

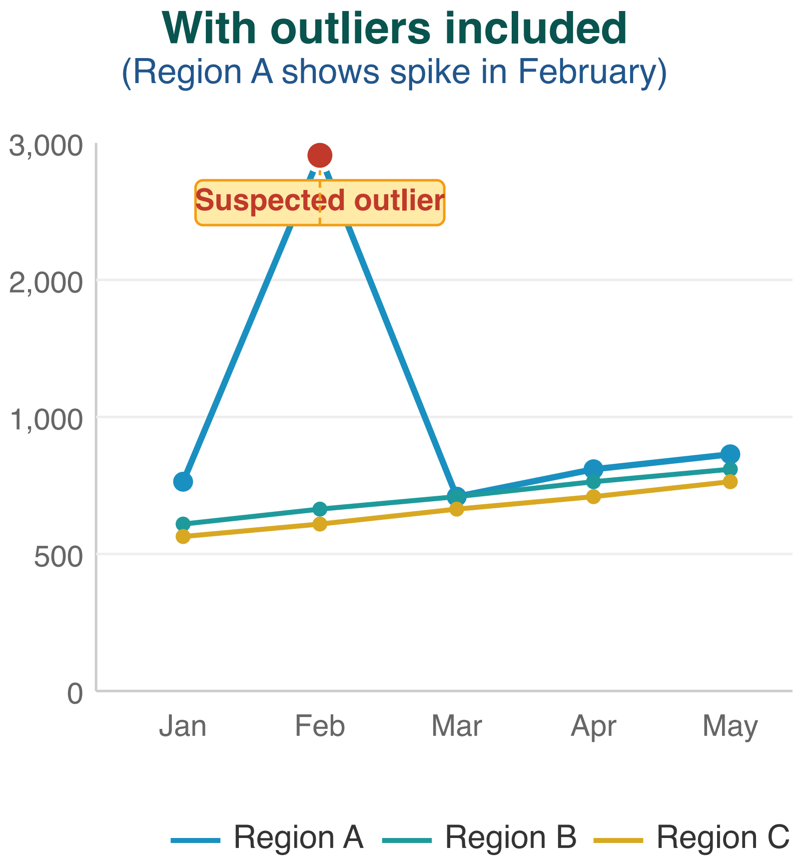
The presence of outliers examines whether a data point in a series of values is extreme (either abnormally high or low) in relation to others in the series.

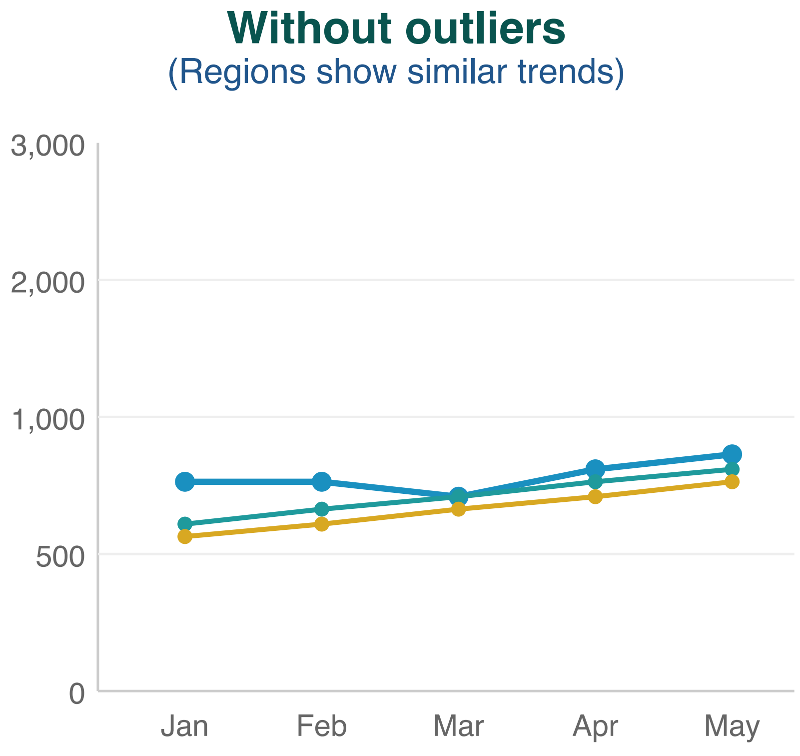
Outliers can be the result of changes in programmatic activities (such as an intensified campaign) or can be data quality problems.

For the FASTR analysis, we identify outliers which are suspiciously high values compared to the usual volume of services reported by the facility (e.g., low values are not identified as outliers in the FASTR analysis).

Outlier illustration

Region A displays an anomalous spike in February that substantially exceeds values reported by other regions — indicative of a data entry error or reporting issue.





Outliers can distort the true picture of service trends

Outlier detection methodology

Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator.

An outlier is defined as:

A value greater than **10 times the median absolute deviation (MAD)** from the monthly median value for the indicator in each time period, **OR** a value for which the proportional contribution in volume for a facility, indicator, and time period is **greater than 80%**

AND for which:

- The volume is **greater than or equal to the median**
- The volume is **not missing**
- The volume is **greater than 100**

Outliers: Percent of monthly values that are outliers

For a given indicator in a given time period, the percent of monthly values that are outliers:

% outliers = # monthly values that are outliers / total N of monthly values

Outliers

Percentage of facility-months that are outliers, May 2024 to Apr 2025

	Antenatal care 1	Antenatal care 4	Institutional delivery	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG vaccine	Penta vaccine 1	Penta vaccine 3	Outpatient visit
Region 001	0.6%	0.2%	0.2%	1.9%	1.2%	0.6%	0.9%	0.6%	0.9%
Region 002	0.6%	0.0%	1.2%	1.0%	1.8%	0.2%	0.2%	0.1%	2.7%
Region 003	0.5%	0.4%	0.1%	0.0%	0.1%	0.1%	0.4%	0.6%	1.4%
Region 004	0.4%	0.0%	0.0%	0.8%	0.0%	0.5%	0.8%	0.9%	0.1%
Region 005	0.5%	0.9%	0.5%	0.8%	0.5%	0.6%	0.8%	0.3%	3.3%
Region 006	0.4%	0.0%	0.1%	0.3%	0.1%	1.3%	2.3%	2.5%	0.6%

3% or above
1% to 2%
Below 1%

Outliers are reports which are suspiciously high compared to the usual volume reported by the facility in other months. Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator. Outliers are defined observations which are greater than 10 times the median absolute deviation (MAD) from the monthly median value for the indicator in each time period, OR a value for which the proportional contribution in volume for a facility, indicator, and time period is greater than 80%. Outliers are only identified for indicators where the volume is greater than or equal to the median, the volume is not missing, and the average volume is greater than 100.

Consistency between related indicators

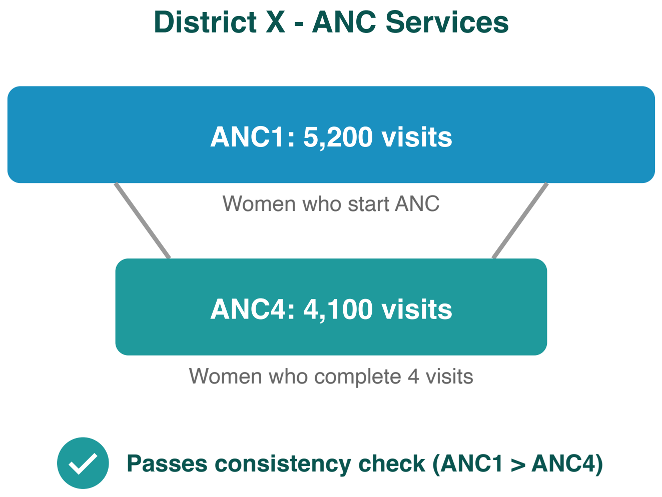
Program indicators with a predictable relationship are examined to determine whether the expected relationship exists between them. In other words, this process examines whether the observed relationship between the indicators, as shown in the reported data, is that which is expected.

Indicator pairs assessed

Indicator pair	Expected relationship
ANC1 / ANC4	Ratio should be ≥ 0.95
Penta1 / Penta3	Ratio should be ≥ 0.95
BCG / Facility delivery	Within 30% (≥0.7 and ≤1.3)

These pairs have expected relationships. We expect ANC1 > ANC4 since not all women complete four visits.

BCG is a birth dose vaccine so we expect similar numbers to facility deliveries, with a 30% tolerance for variability.

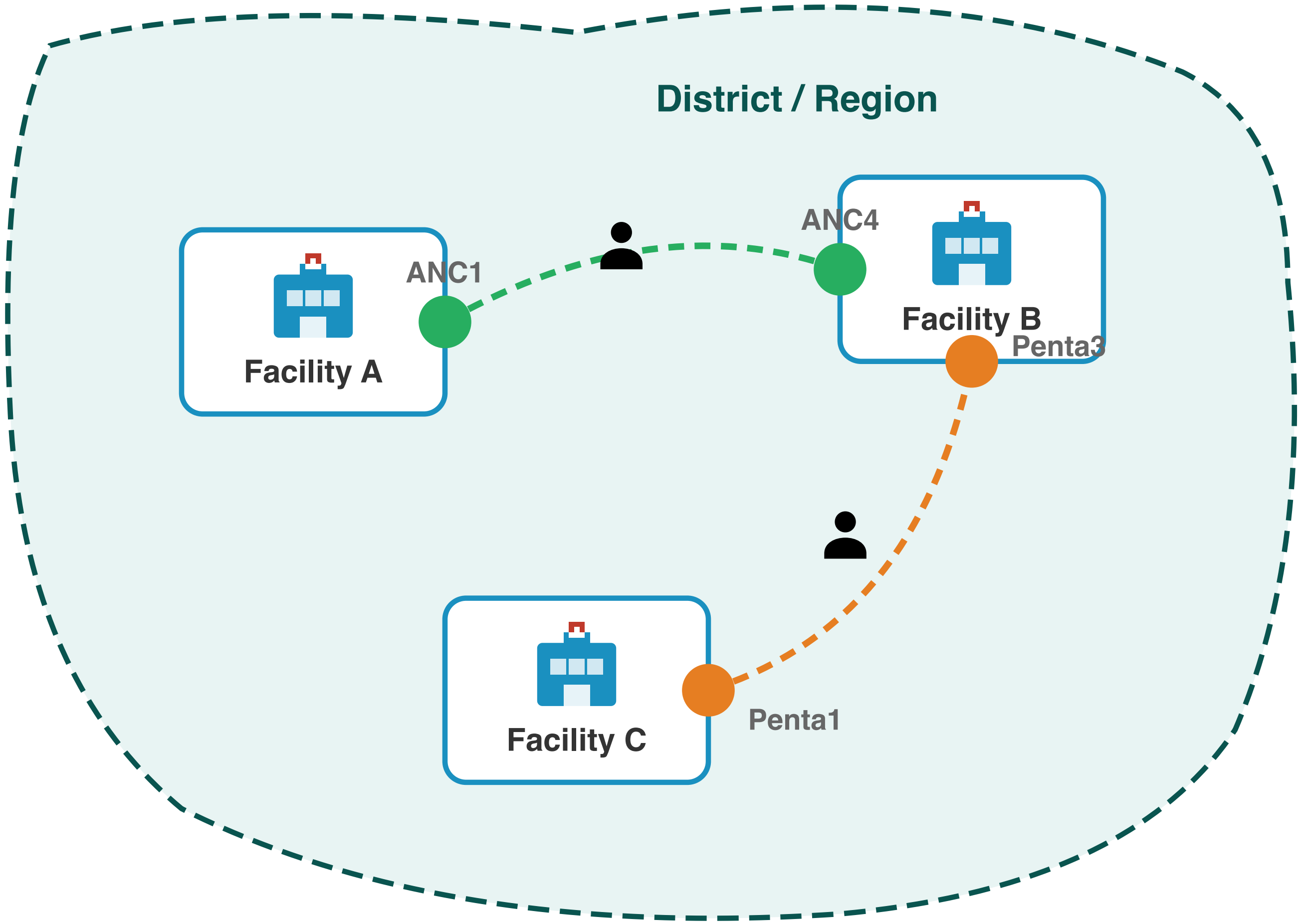


Why assess consistency at district level?

Patients often access different services at different facilities within a district:

- A woman may attend **ANC1** at a nearby health post, but travel to a health centre for **ANC4**
- A child may receive **Penta1** at a local clinic, but complete **Penta3** at a district hospital

Checking consistency at the facility level would miss these patterns. Aggregating to district level captures the complete picture of service utilization within a geographic area.



Internal consistency: FASTR output

Internal consistency

Percentage of sub-national areas meeting consistency benchmarks, May 2024 to Apr 2025

	ANC1 is larger than ANC4	Delivery is approximately equal to BCG	Penta1 is larger than Penta3
Region 001	100.0%	0.0%	97.2%
Region 002	100.0%	0.0%	100.0%
Region 003	100.0%	45.8%	83.3%
Region 004	100.0%	37.5%	87.5%
Region 005	100.0%	0.0%	87.5%
Region 006	100.0%	49.0%	84.4%

90% or above
80% to 89%
Below 80%

Internal consistency assesses the plausibility of reported data based on related indicators. Consistency metrics are approximate - depending on timing and seasonality, indicator definitions, and the nature of service delivery and reporting, values may be expected to sit outside plausible ranges. Indicators which are similar are expected to have roughy the same volume over the year (within a 30% margin). The data in this analysis is adjusted for outliers.

Data quality summary score

A composite measure of data quality provides an overall view of how well a dataset meets quality standards.

By integrating multiple dimensions of data quality into a single score, it simplifies the interpretation of detailed information from several measures. This allows health systems to quickly assess the reliability of data, making it easier to identify trends and issues at a glance.

Definition of adequate data quality

For the FASTR analysis, we defined adequate data quality as:

- No missing indicator data for OPD, Penta1, and ANC1, where available, **AND**
- No outliers for OPD, Penta1, and ANC1, where available, **AND**
- Consistent reporting between Penta1/Penta3 and ANC1/ANC4

Overall DQA score: Percent of monthly values meeting all criteria

For a given indicator in a given time period, the percent of monthly values meeting all DQA criteria:

% adequate quality = # monthly values meeting all criteria / total N of monthly values

Overall DQA score

Percentage of facility-months with adequate data quality over time

	2022	2023	2024	2025
Region 001	60.8%	76.6%	76.4%	84.4%
Region 002	55.3%	61.2%	58.2%	52.0%
Region 003	69.3%	73.4%	60.6%	47.0%
Region 004	54.6%	70.7%	70.0%	84.9%
Region 005	57.9%	69.5%	56.6%	55.4%
Region 006	74.0%	84.7%	72.2%	71.7%

80% or above
70% to 79%
Below 70%

Adequate data quality is defined as: 1) No missing data or outliers for OPD, Penta1, and ANC1, where available 2) Consistent reporting between Penta1/Penta3 and ANC1/ANC4.

Mean DQA score: How close are we to adequate quality?

The mean DQA score shows how close a facility's data is to meeting all quality criteria. A score of **100% means the data passes** all DQA checks—no missing values, no outliers, and consistent reporting.

Mean DQA = (completeness & outlier score + consistency score) / 2

Mean DQA score

Average data quality score across facility-months

	2022	2023	2024	2025
Region 001	90.2%	95.5%	95.5%	96.5%
Region 002	87.3%	89.0%	88.2%	85.6%
Region 003	92.8%	94.8%	91.0%	84.3%
Region 004	88.0%	93.9%	93.0%	97.0%
Region 005	89.5%	93.6%	91.1%	88.6%
Region 006	93.2%	96.3%	93.4%	93.4%

80% or above
70% to 79%
Below 70%

Items included in the DQA score include: No missing data for 1) OPD, 2) Penta1, and 3) ANC1, where available; No outliers for 4) OPD, 5) Penta1, and 6) ANC1, where available; Consistent reporting between 7) Penta1/Penta3, 8) ANC1/ANC4, 9)BCG/Delivery, where available.

☕ Tea Break

15 minutes

Back at

Data quality adjustment - Module 2

Correcting outliers and imputing missing values to improve data reliability

Rationale for data quality adjustment

Routine HMIS data contain two common limitations that can distort analytical results:

Issue	Impact on analysis
Outliers	Extreme values create artificial spikes in service volumes
Incomplete reporting	Missing data creates artificial declines that do not reflect actual service delivery

FASTR addresses these limitations by replacing problematic values with estimates derived from each facility's historical reporting patterns.

Adjustment scenarios

To support transparency and sensitivity analysis, FASTR produces four parallel datasets:

Scenario	Description
Unadjusted	Original reported values
Outliers adjusted	Extreme values replaced
Completeness adjusted	Missing values imputed
Both adjusted	All corrections applied

Indicators excluded from adjustment

Certain indicators are excluded from the adjustment process:

- **Mortality indicators** (maternal deaths, neonatal deaths, under-5 deaths): These represent discrete events where smoothing or imputation is not appropriate
- **Low-volume indicators**: Indicators that never exceed 100 reported events in any month are excluded from adjustment

Outlier adjustment methodology

Outlier values are replaced using facility-specific historical data. The adjustment follows a hierarchical approach:

Priority	Method	Application
1	Centered 6-month average	3 months before + 3 months after the outlier
2	Forward 6-month average	When insufficient preceding data (e.g., start of series)
3	Backward 6-month average	When insufficient following data (e.g., end of series)
4	Same month, previous year	When rolling averages unavailable; useful for seasonal indicators
5	Facility historical mean	Mean of all valid values for this indicator at this facility

Outlier adjustment: FASTR output

Deviance Due to Outliers

Percent change in volume due to outlier adjustment, Oct 2024 to Sep 2025

	Antenatal client 1st visit	Antenatal client 4th visit	Institutional delivery (normal, assisted, and csection)	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG dose	Pentavalent 1st dose	Pentavalent 3rd dose	Family planning methods - long acting	Family planning methods - short acting	Fever case tested positive for malaria (RDT and microscopy)	Malaria treated with ACT
Bo District	0.0%	0.2%	4.6%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.2%	0.2%
Bombali District	1.3%	0.0%	0.0%	2.3%	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	0.0%	0.0%
Bonthe District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	2.5%	0.0%	0.0%	0.0%
Falaba District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.7%	0.0%	0.0%	2.2%	0.0%	0.0%
Kailahun District	0.0%	0.0%	0.6%	0.0%	0.0%	0.0%	0.0%	0.0%	4.5%	2.1%	0.2%	0.2%
Kambia District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	0.0%	0.0%	0.0%
Karene District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	2.7%	1.3%	5.7%	4.7%	0.0%	0.0%
Kenema District	0.0%	0.0%	0.0%	0.0%	0.0%	1.2%	1.7%	1.4%	1.8%	0.4%	0.0%	0.0%
Koinadugu District	0.0%	0.0%	0.0%	0.0%	0.3%	0.0%	0.0%	0.0%	2.6%	1.9%	0.0%	0.0%
Kono District	0.7%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	3.2%	0.6%	2.0%	1.8%
Moyamba District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	3.7%	0.4%	0.1%	0.1%
Port Loko District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.8%	0.0%	0.0%	0.0%
Pujehun District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.3%	0.3%
Tonkolili District	0.0%	0.3%	0.0%	0.0%	0.0%	0.6%	0.0%	0.0%	1.5%	1.4%	0.1%	0.1%
Western Area Rural District	0.0%	0.0%	0.7%	1.5%	1.4%	0.5%	1.0%	0.0%	0.0%	0.0%	0.0%	0.0%
Western Area Urban District	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.6%	0.9%	1.1%	0.5%	1.5%	0.3%

3% or above
1% to 2%
Below 1%

Outliers are reports which are suspiciously high compared to the usual volume reported by the facility in other months. Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator. Outliers are defined observations which are greater than 10 times the median absolute deviation (MAD) from the monthly median value for the indicator in each time period, OR a value for which the proportional contribution in volume for a facility, indicator, and time period is greater than 80%. Outliers are only identified for indicators where the volume is greater than or equal to the median, the volume is not missing, and the average volume is greater than 100. The deviance is the difference in volume after removing the outlier. High levels of deviance can affect the plausiability of the data.

Completeness adjustment methodology

For months identified as incomplete or missing, values are imputed using the same 6-month rolling average approach applied to outlier adjustment.

Priority	Method	Application
1	Centered 6-month average	When sufficient data exists before and after the gap
2	Forward 6-month average	For gaps at the start of the time series
3	Backward 6-month average	For gaps at the end of the time series
4	Facility historical mean	Mean of all valid values for this indicator at this facility

This approach prevents temporary reporting gaps from creating artificial declines in service volumes.

Completeness adjustment: FASTR output

Deviance Due to Incompleteness

Percent change in volume due to completeness adjustment, Oct 2024 to Sep 2025

	Antenatal client 1st visit	Antenatal client 4th visit	Institutional delivery (normal, assisted, and csection)	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG dose	Pentavalent 1st dose	Pentavalent 3rd dose	Family planning methods - long acting	Family planning methods - short acting	Fever case tested positive for malaria (RDT and microscopy)	Malaria treated with ACT
Bo District	2.2%	4.7%	3.1%	4.7%	3.8%	2.2%	7.2%	9.0%	12.3%	18.4%	3.6%	8.3%
Bombali District	2.6%	3.3%	4.1%	10.4%	12.9%	7.4%	5.9%	5.6%	13.0%	5.6%	10.8%	8.6%
Bonthe District	1.5%	5.2%	3.6%	8.3%	4.3%	1.6%	4.1%	4.0%	24.3%	8.3%	2.7%	6.1%
Falaba District	0.5%	2.8%	3.0%	6.4%	4.2%	0.9%	0.3%	0.3%	12.8%	7.5%	1.8%	4.7%
Kailahun District	0.5%	1.9%	1.3%	13.8%	11.9%	0.5%	0.4%	0.4%	8.3%	8.1%	2.8%	4.4%
Kambia District	0.8%	2.8%	0.4%	1.4%	1.3%	0.5%	0.6%	0.8%	9.6%	5.3%	0.3%	1.4%
Karene District	0.3%	0.8%	0.0%	0.0%	0.0%	0.0%	0.0%	0.3%	8.3%	6.3%	2.3%	4.8%
Kenema District	0.3%	1.2%	0.7%	1.7%	1.3%	0.1%	0.2%	0.2%	5.6%	4.5%	0.5%	1.1%
Koinadugu District	2.3%	3.9%	2.7%	4.3%	2.6%	1.5%	2.1%	2.3%	41.1%	23.9%	13.1%	14.9%
Kono District	0.4%	2.1%	1.2%	5.6%	3.1%	0.6%	0.7%	0.9%	6.8%	7.3%	1.2%	6.5%
Moyamba District	1.4%	3.8%	2.9%	4.0%	3.3%	1.7%	2.0%	2.2%	13.9%	7.4%	2.8%	5.3%
Port Loko District	1.5%	2.0%	2.1%	2.9%	4.8%	1.6%	1.3%	1.4%	19.9%	9.8%	10.3%	8.8%
Pujehun District	0.3%	1.1%	2.1%	5.2%	4.0%	0.1%	0.2%	0.3%	7.0%	3.9%	0.3%	0.9%
Tonkolili District	1.2%	3.0%	4.6%	6.5%	2.7%	1.5%	1.8%	1.7%	9.6%	8.2%	2.5%	3.7%
Western Area Rural District	0.7%	0.8%	1.7%	1.7%	2.0%	0.6%	2.6%	2.4%	22.7%	10.1%	0.2%	0.6%
Western Area Urban District	6.2%	8.6%	5.7%	15.9%	13.5%	5.7%	6.6%	6.4%	19.9%	15.5%	4.5%	6.4%

3% or above
1% to 2%
Below 1%

Completeness is defined as the percentage of reporting facilities each month out of the total number of facilities expected to report. A facility is expected to report if it has reported any volume for each indicator anytime within a year. The deviance is the difference in volume after imputing incomplete data. High levels of deviance can affect the plausiability of the data.

☕ Tea Break

15 minutes

Back at

See You Tomorrow!

Day 2 Complete

We resume tomorrow at **09:00**

Service utilization analysis - Module 3

Detecting and quantifying changes in health service delivery over time

Objectives

1. Measure changes in service volume

Year-over-year comparison of service volumes identifies increases or decreases across regions and indicators.

2. Detect and quantify disruptions

Statistical comparison of observed volumes against expected levels—derived from historical trends and seasonal patterns—enables identification and quantification of service shortfalls or surpluses.

Service utilization over time

Service utilization over time

Jan 2019 to Sep 2025

Service volume is adjusted for outliers and completeness.

Year-over-year change

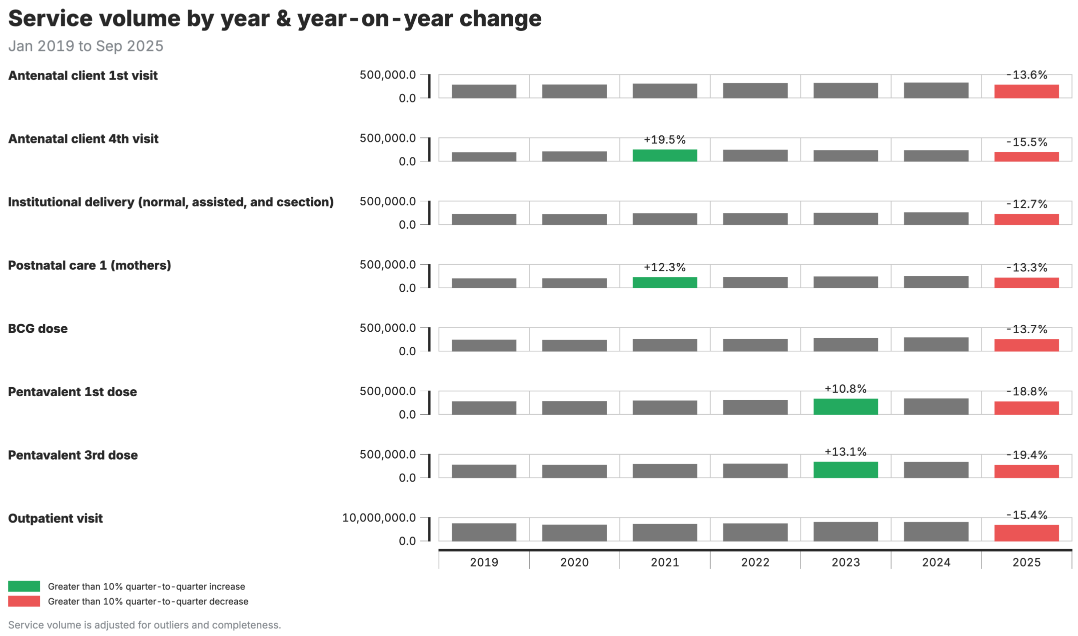
Year-over-year percent change quantifies shifts in service delivery between consecutive years.

For each indicator and region, total volume in the current year is compared to the previous year:

Percent change = (Current year - Previous year) ÷ Previous year × 100

Changes exceeding **±10%** are flagged for review.

Output: Change in service volume



Interpretation

Bars show annual service volumes by region. **Percentages** indicate year-over-year change.

Key considerations:

- Which regions exhibit the largest changes?
- Are changes consistent across regions or geographically concentrated?
- Do patterns vary by indicator?

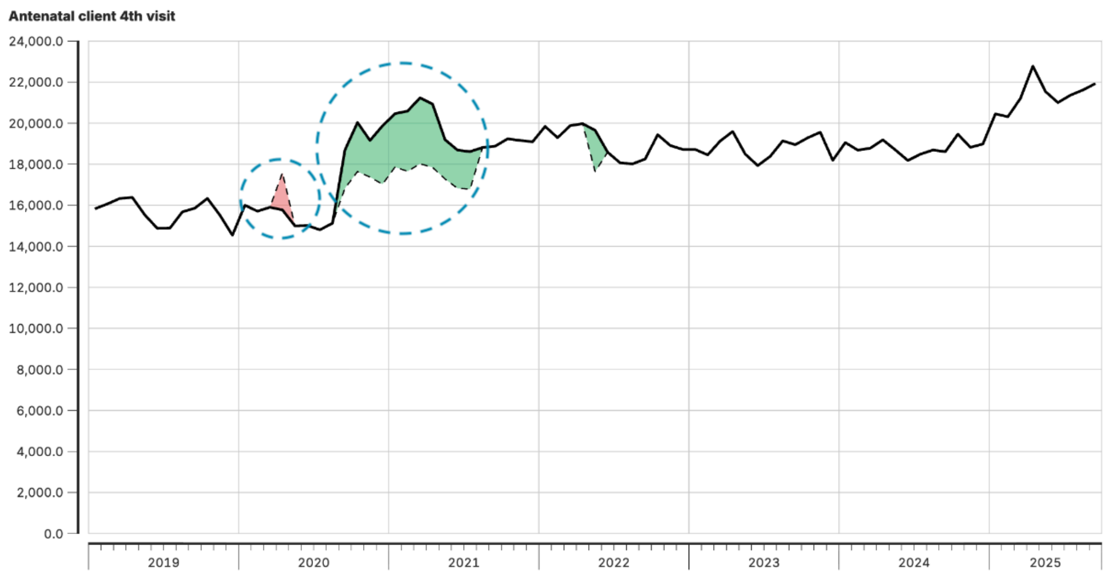
Disruption detection

Our approach to service disruptions and surpluses utilizes an interrupted time series regression with facility-level fixed effects. Previous large and unexpected changes in historical data are removed. Unexpected volume changes are estimated by comparing observed volume to expected volume based on historical trends and seasonality.

Disruptions and surpluses

Disruptions are flagged when volumes fall below anticipated levels, signaling potential barriers to access, resource shortages, or system failures.

Surpluses occur when volumes exceed expectations, which may indicate increased demand, over-reporting, or changes in service delivery.



How it works

Using past data to set expectations: We look at the past few years of data to understand the typical pattern for each month, accounting for regular seasonal changes.

Spotting unusual changes: We compare current service volumes to expectations. If we see volumes much higher or lower than expected, we flag it as an unusual change.

Handling past disruptions: We adjust historical data by removing previous big, unexpected changes so one-off events don't skew our understanding of what's "normal."

Detecting disruptions over time: We look at trends to see if there are clear shifts in health service use over several months.

Comparison to DHIS2

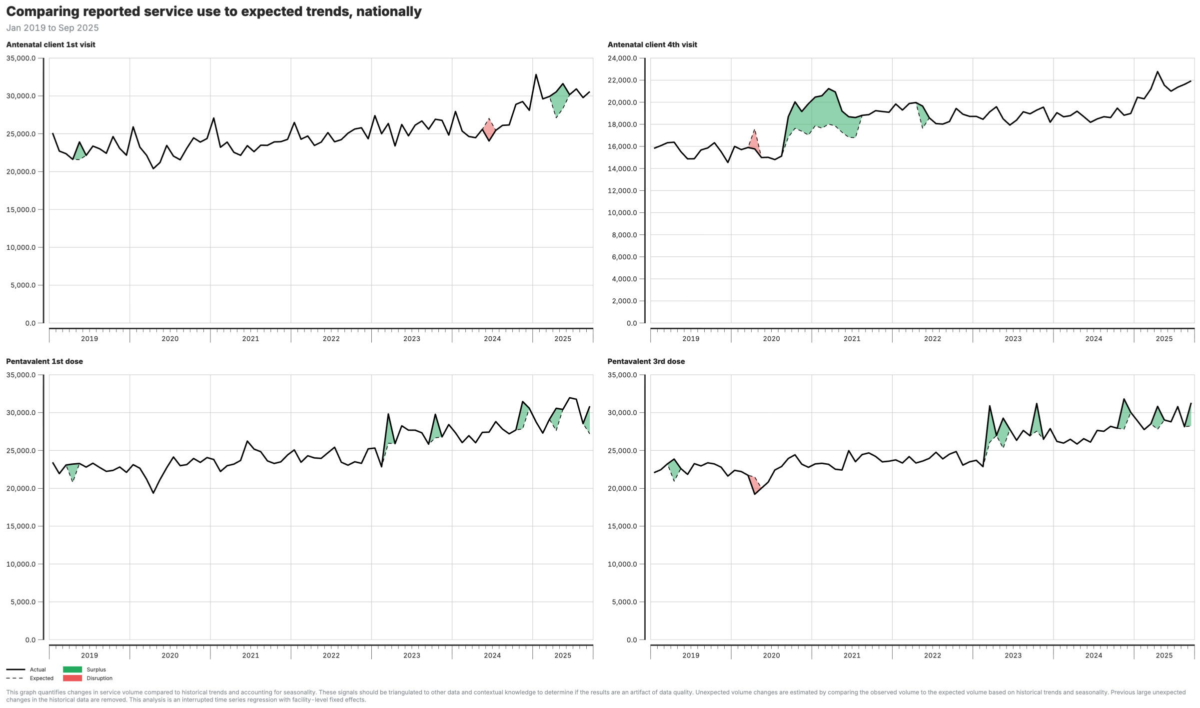
Extension of service utilization analysis, using more complex statistical approaches not available in DHIS2.

Using a regression framework, we are able to:

- Account for seasonality
- Exclude unusual changes to ensure one-off events aren't influencing normal trends
- Use historical data as a baseline for context
- Detect disruptions and recovery patterns
- Quantify changes with a robust methodology as compared to just observing simple fluctuations in a trend line

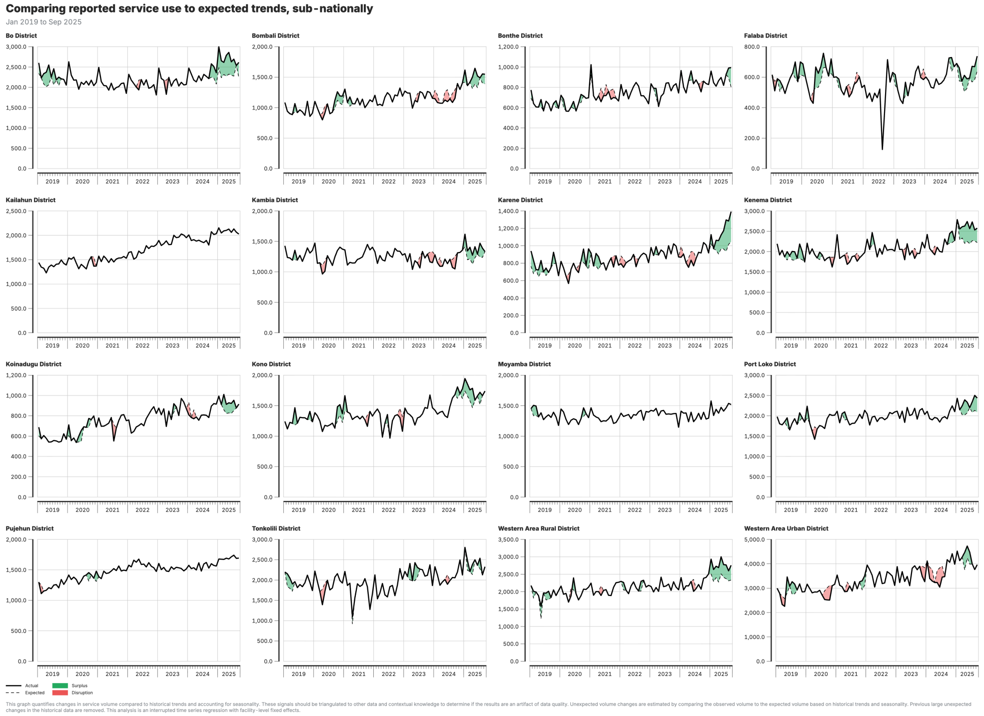
This improves the ability to interpret and compare utilization data across national and sub-national areas without needing population denominators.

Output: Actual vs expected (national)



National-level comparison of observed service volumes against expected values derived from historical trends and seasonal patterns.

Output: Actual vs expected (subnational)



Subnational disaggregation enables identification of geographic areas where disruptions are concentrated.

Service coverage estimates - Module 4

Estimating the percentage of the target population that received a given health service

Our approach to service coverage analysis

Our approach derives and validates population denominators, significantly improving coverage estimates reported from HMIS systems.

In countries with accurate data, this approach helps identify subnational inequities and updates outdated estimates, while in countries with less precise data, the trends still provide valuable insights into performance.

We use these estimates to track recent trends and subnational disparities in the coverage of selected health services.

Two-part analytical process

The coverage estimation module operates in two sequential parts:

Part	Components
Part 1: Denominator calculation	Calculate target populations using multiple methods; compare against survey benchmarks; select optimal denominator for each indicator
Part 2: Coverage estimation	Apply denominator selections; project survey estimates forward using HMIS trends; generate final coverage estimates

What is service coverage?

Service coverage represents the proportion of the target population that received a specified health service.

$$\text{Service coverage} = \frac{\text{Population who received the service}}{\text{Population who need the service (target population)}}$$

Numerator comes from DHIS2 data, count of services adjusted for outliers

Several options for estimating the denominator

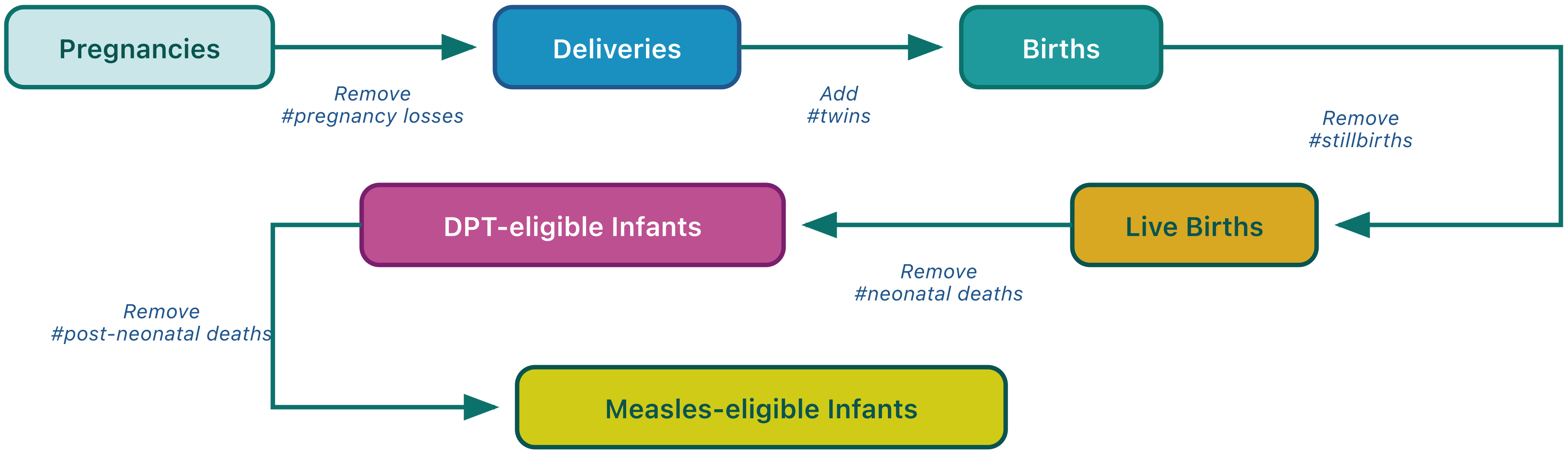
Denominators by service type

Each health indicator corresponds to a specific target population:

Service	Target population (denominator)
ANC1, ANC4	Pregnant women
Institutional delivery	Live births
BCG	Live births
Penta1, Penta3	Infants surviving beyond neonatal period
Measles1, Measles2	Infants surviving beyond infancy

Demographic cascade

Sequential demographic adjustments transform one target population estimate into another. Starting from pregnancies, demographic factors are applied to derive subsequent denominators:



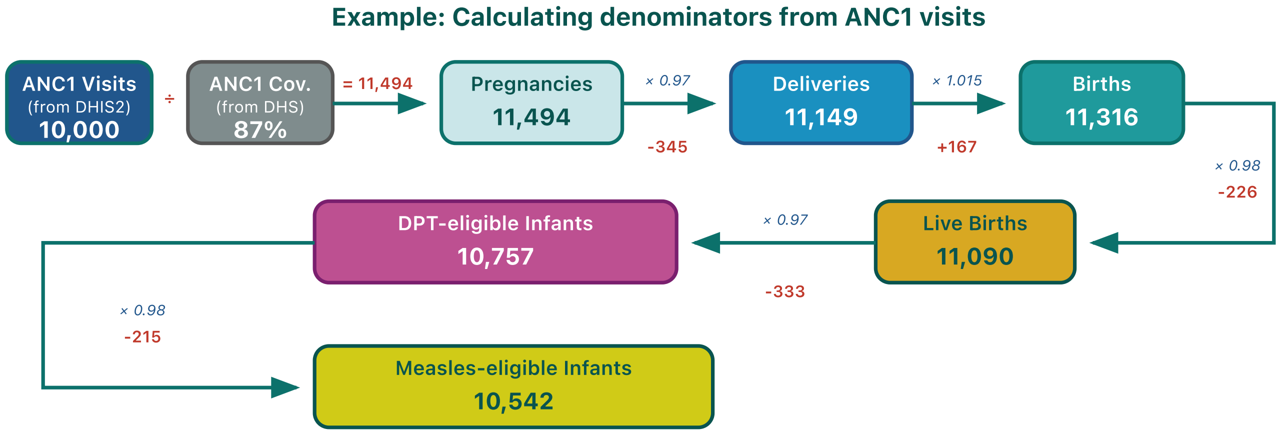
Default values:

Pregnancy loss rate = 0.03 | Twin rate = 0.015 | Stillbirth rate = 0.02 | NMR = 0.03 | PNMR = 0.02

Use country-specific values for stillbirth rate, NMR, PNMR, and IMR when available from DHS/MICS.

Denominator cascade: Illustration

Starting from ANC1 service counts, demographic adjustment factors are applied sequentially to derive denominators for other services:



Calculation summary:

Step 1: Pregnancies = ANC1 visits ÷ ANC1 coverage = 10,000 ÷ 0.87 = 11,494

Step 2: Apply demographic factors through the cascade:

Pregnancies (11,494) → Deliveries (11,149) → Births (11,316) → Live Births (11,090) → DPT-eligible (10,757) → Measles-eligible (10,542)

Rates: Pregnancy loss = 3%, Twin rate = 1.5%, Stillbirth = 2%, NMR = 3%, PNMR = 2%

Forward and backward derivation

From any entry point, the cascade derives denominators in both directions:

Direction	Method	Example from Penta1
Forward	Apply mortality/attrition rates	DPT-eligible → Measles1-eligible → Measles2-eligible
Backward	Reverse mortality rates (add deaths back)	DPT-eligible → Live births → Births → Deliveries → Pregnancies

Backward derivation enables estimation of upstream populations from downstream service counts.

Denominators by entry point

Each HMIS indicator serves as an entry point. The module derives all target populations via forward and backward cascades:

Entry point	Base calculation	Forward derivation	Backward derivation
ANC1	ANC1 ÷ coverage → Pregnancies	Deliveries → Live births → DPT-eligible → Measles-eligible	—
Deliveries	Deliveries ÷ coverage → Deliveries	Live births → DPT-eligible → Measles-eligible	Pregnancies
BCG	BCG ÷ coverage → Live births	DPT-eligible → Measles-eligible	Deliveries → Pregnancies
Penta1	Penta1 ÷ coverage → DPT-eligible	Measles1-eligible → Measles2-eligible	Live births → Births → Deliveries → Pregnancies
UN WPP	Crude birth rate × population → Pregnancies, live births; Under-1 pop → DPT, measles	Applies mortality rates for measles denominators	—

Automatic denominator selection

For each indicator, the module selects the denominator that produces coverage closest to the survey benchmark.

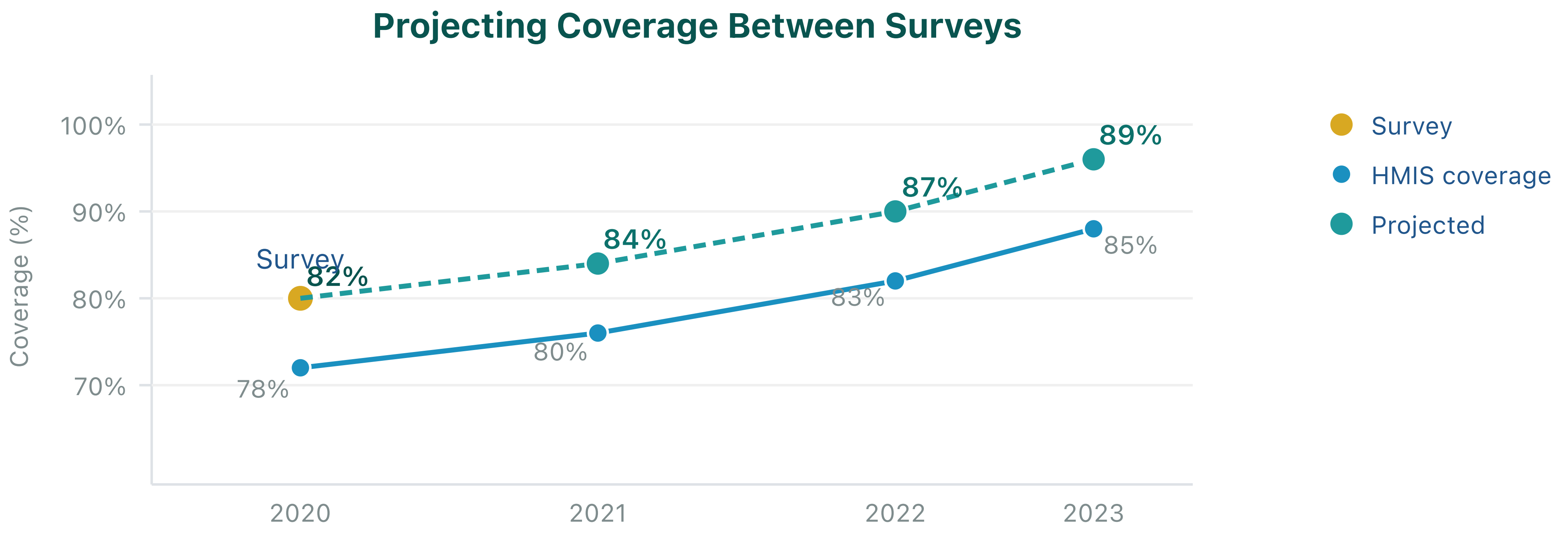
Selection algorithm:

1. Calculate coverage using each denominator option
2. Calculate squared error against survey: $(coverage - survey)^2$
3. Apply selection hierarchy (HMIS-based denominators prioritized over UN WPP)
4. Select the HMIS-based denominator with minimum error

Selection is made per indicator and geographic area. Users may override automatic selections in Part 2.

Coverage projection methodology

The module projects the most recent survey value forward using trends observed in HMIS-derived coverage:



Year-over-year changes (deltas) in HMIS coverage are calculated and applied to the last survey value. This approach preserves the survey baseline while incorporating observed service delivery trends.

Interpretation of coverage outputs

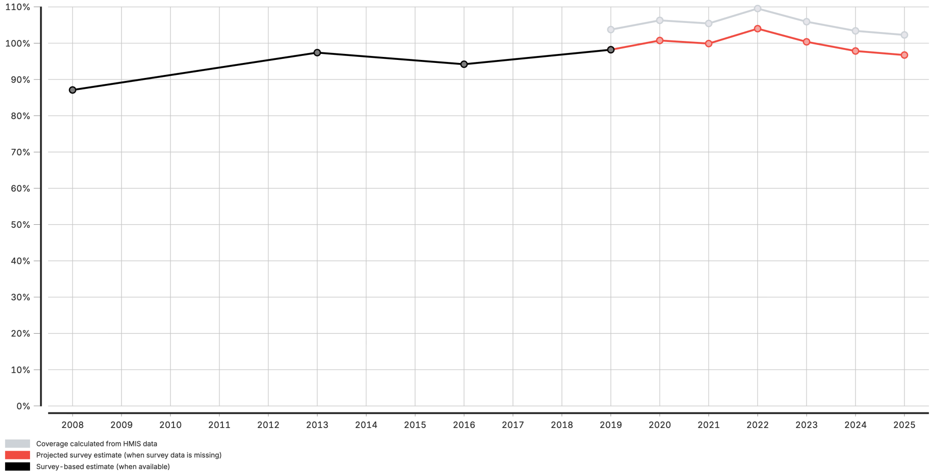
Element	Description
Black line/points	Survey data (DHS/MICS) — household survey reference
Grey line/points	HMIS-based coverage from facility data
Red line/points	Projected coverage — survey estimates extended using HMIS trends

Coverage (national)

Coverage estimates for Antenatal client 1st visit

2008 to 2025

DISCLAIMER: These results use routine data to provide rigorous, but not official estimates. They should be interpreted considering any data quality or representation limitations, including data quality findings and any other country specific factors.



Estimating service coverage from administrative data can provide more timely information on coverage trends, or highlight data quality concerns. Numerators are the volumes reported in HMIS, adjusted for data quality. Denominators are selected from UN projections, survey estimates, or derived from HMIS volume for related indicators. National projections are made by applying HMIS trends to the most recent survey data. Subnational estimates are more sensitive to poor data quality, and projections from surveys are not calculated.

MICS data courtesy of UNICEF. Multiple Indicator Cluster Surveys (various rounds) New York City, New York.

DHS data courtesy of ICF. Demographic and Health Surveys (various rounds). Rockville, Maryland.

Data for the current year reflect the period available at the time of analysis; population figures have been scaled to match the corresponding duration.

Interpretation

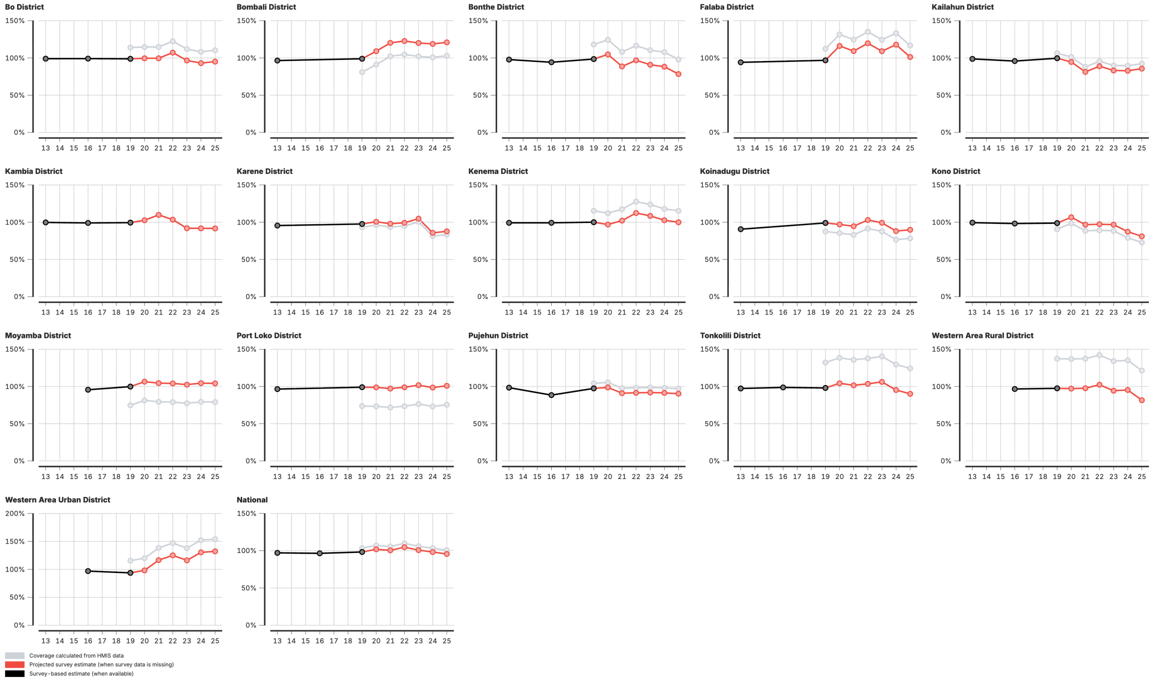
Black line/points show survey data (DHS/MICS) as the household survey reference. **Grey line/points** show HMIS-based coverage from facility data. **Red line/points** show projected coverage — survey estimates extended using HMIS trends.

Coverage (subnational)

Subnational coverage estimates for Antenatal client 1st visit

2013 to 2025

DISCLAIMER: These results use routine data to provide rigorous, but not official estimates. They should be interpreted considering any data quality or representation limitations, including data quality findings and any other country specific factors.



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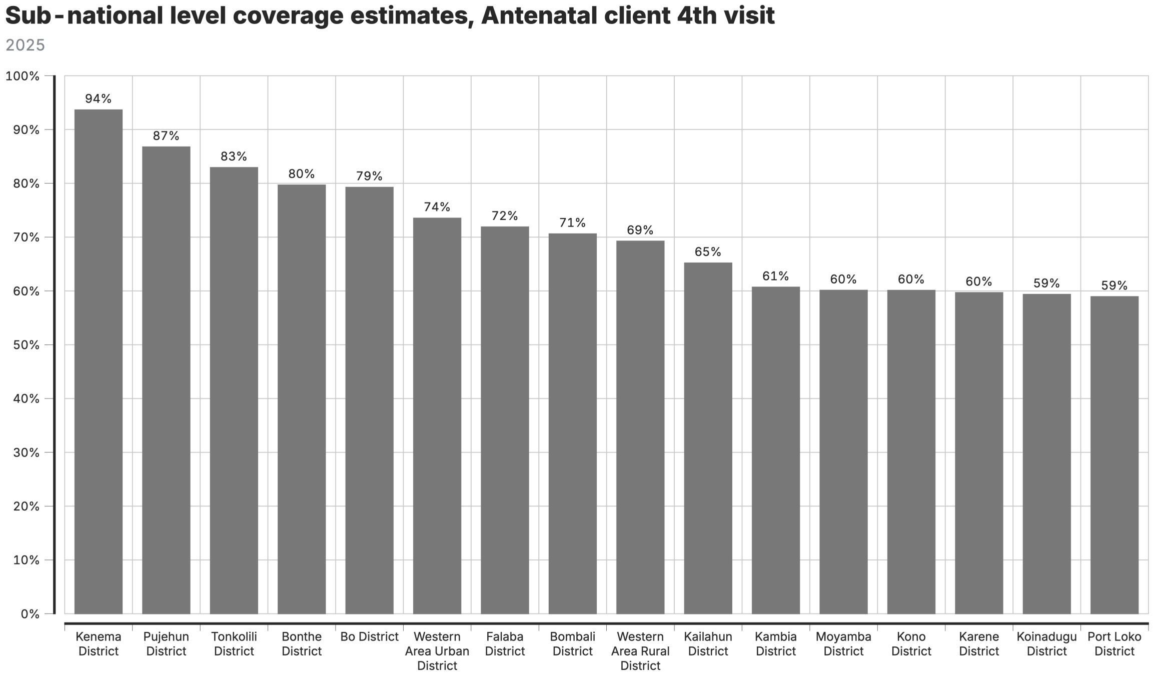
Coverage (subnational)



Estimating service coverage from administrative data can provide more timely information on coverage trends, or highlight data quality concerns. Numerators are the volumes reported in HMIS, adjusted for data quality. Denominators are selected from UN projections, survey estimates, or derived from HMIS volume for related indicators. National projections are made by applying HMIS trends to the most recent survey data. Subnational estimates are more sensitive to poor data quality, and projections from surveys are not calculated.

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Data for the current year reflect the period available at the time of analysis; population figures have been scaled to match the corresponding duration.

🍽️ Lunch Break

60 minutes

Back at

Analytical thinking & interpretation

Interpretation connects **data patterns** to **programmatic meaning**.

For every FASTR output, ask three questions:

1. **What does it show?** — Describe the pattern accurately
2. **Why might that be?** — Consider multiple explanations
3. **What should we do?** — Identify next steps or actions

Moving from numbers to insights requires context, critical thinking, and programmatic knowledge.

Moving from data to key messages

What is a result?

Results are what the analysis found (data quality scores, coverage estimations, service utilization numbers). They are often many in number, complex, hard to understand 'at a glance', and lacking interpretation.

What is a key takeaway?

Key takeaways are what the results are telling us — why the results matter, the 'so what'. They should be few in number, simple and clear, easy to remember, and actionable.

Dissemination and data use roadmap

A data use roadmap is a strategic plan that outlines how data will be utilized, shared, and disseminated effectively.

Why is it important?

- Establishes metrics to evaluate the success of data use
- Identifies key stakeholders, target audiences, and dissemination platforms
- Outlines steps needed to achieve data dissemination goals
- Anticipates potential challenges and develops strategies to solve for them

Presenting reports and group feedback

Content to be developed

Generating quarterly reporting products

Content to be developed

This section will cover:

- Quarterly reporting workflow
- Using the FASTR platform for automated reports
- Quality assurance for reports
- Distribution and feedback mechanisms

☕ Tea Break

15 minutes

Back at

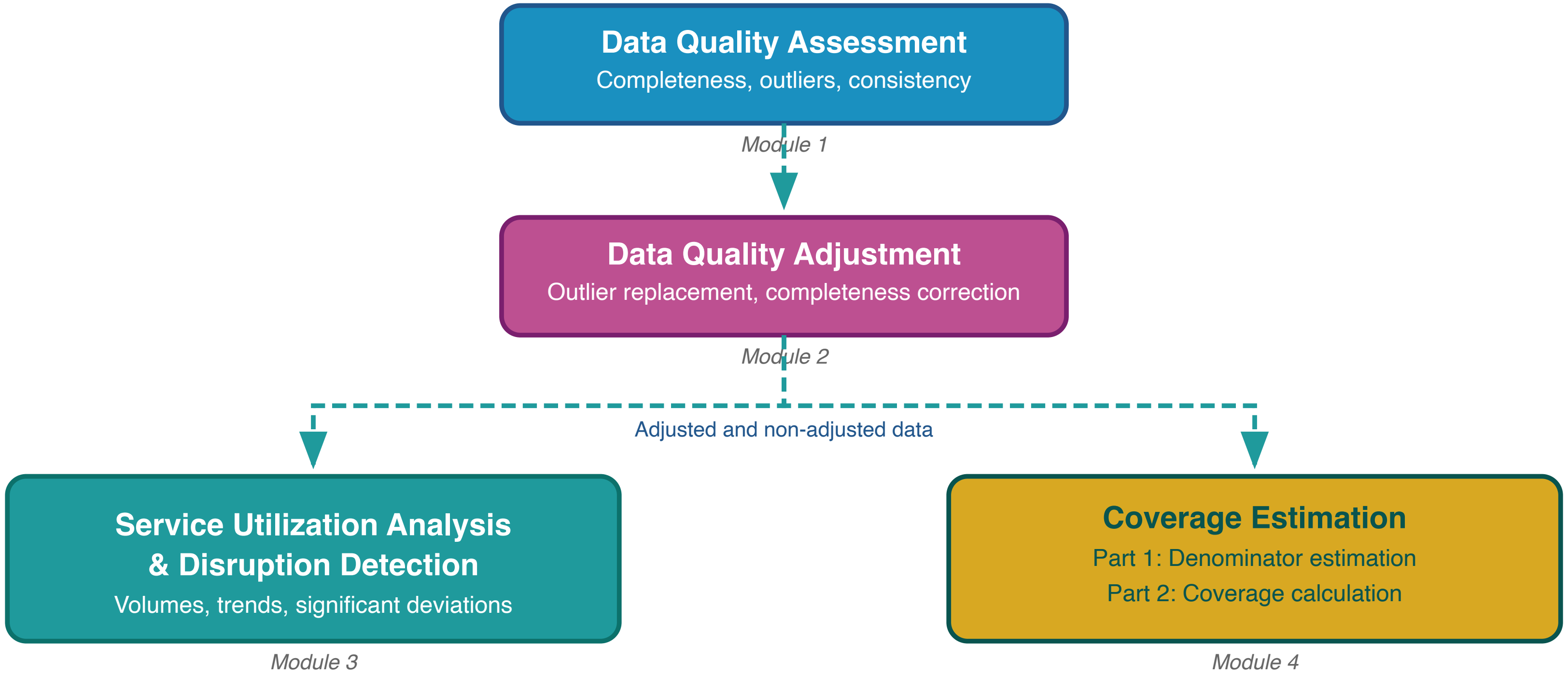
Contact Information

FASTR Team

Email:

Website: https://www.globalfinancingfacility.org/

FASTR analytical pipeline



The FASTR analysis follows a sequential workflow where each step builds on the previous:

1. **Assess data quality** - Identify issues with completeness, outliers, and consistency
2. **Adjust for quality issues** - Apply corrections to improve data reliability
3. **Analyze adjusted data** - Generate service utilization and coverage estimates

FASTR takes a multi-pronged approach to data quality, with the belief that data quality should not be a barrier to data use – with the right feedback loops, use of data can contribute to improved quality

- We do granular data quality assessments and adjustments based on facility-level data leveraging HMIS access with an API
- We use only high-volume indicators, because the most-used services provide more stable estimates
- We focus on variations across time and space rather than specific point estimates and discuss the interpretation and relevance for decision-making with in-country decision-makers
- We believe that using the data and providing feedback is the first step to improving the data

We will discuss each of these areas over the next sessions.

Rationale for data quality assessment

Challenge: Routine health facility data may contain quality limitations:

- Reported values may fall outside plausible ranges
- Reporting gaps affect data completeness
- Inconsistencies exist between related indicators

Implications: Data quality limitations affect decision-making

- Inaccurate assessments of service delivery trends
- Misidentification of areas requiring intervention
- Suboptimal resource allocation

Assessing and adjusting for data quality

Quality assessments identify the highest priority issues and necessary analytical adjustments, so quality issues do not become a barrier to data analysis and use.

Objective 1: Enable analytical adjustment

Systematic data quality assessment supports the application of targeted adjustments, enhancing the utility of HMIS data for evidence-based decision-making.

Objective 2: Monitor data quality trends

Data quality assessment enables ongoing monitoring to:

- Inform indicator selection based on quality profiles across the HMIS
- Guide targeted data quality interventions and supportive supervision in areas with weaker data quality
- Evaluate the effectiveness of data quality improvement initiatives over time

Measures of data quality

Measures of data quality

Measures of data quality - detailed (1/2)

Domain	What does it measure?	How is it assessed?
Completeness	Are all data present?	Reporting completeness: whether all units report. Indicator completeness: whether values are recorded for specific data elements
Timeliness	Are data regularly submitted on time?	Whether units submitted reports before the set deadline

Measures of data quality - detailed (2/2)

Domain	What does it measure?	How is it assessed?
Consistency	Are data plausible in view of what has been previously reported?	Presence of outliers, consistency over time, consistency between related indicators, external comparison with other data sources, consistency of population data
Accuracy	Do data faithfully reflect actual service delivery?	Review of source documents and comparison to monthly reports and HMIS values (data verification factor)

How does FASTR data quality analysis differ from DHIS2? (1/2)

Purpose of data quality assessment

- **DHIS2:** focuses on data quality assessment to routinely strengthen data quality over time
- **FASTR:** focuses on assessing data quality to inform an analysis which answers a pressing policy question

Data quality adjustment

- **DHIS2:** focus is on identifying data quality issues and working with facilities to improve reporting practices
- **FASTR:** focus on applying analytical adjustment techniques to account for data quality issues in the analysis; goal is to generate the most robust estimates despite data quality challenges

How does FASTR data quality analysis differ from DHIS2? (2/2)

Selection of indicators, measures, and thresholds – FASTR focuses on DQA elements most relevant for analysis

- The purpose of the data quality assessment guides the selection of indicators, measures, and thresholds
- DHIS2 allows configuration of a DQA dashboard for any selection of indicators; FASTR selects indicators that will be used in a specific analysis
- DHIS2 DQA includes timeliness as a measure of data quality. FASTR does not include timeliness - it is important for strengthening routine reporting but less important for analysis with available data
- DHIS2 DQA includes reporting completeness and indicator completeness while FASTR focuses only on indicator completeness

Indicator completeness

What it measures: The extent to which facilities report data on selected core indicators

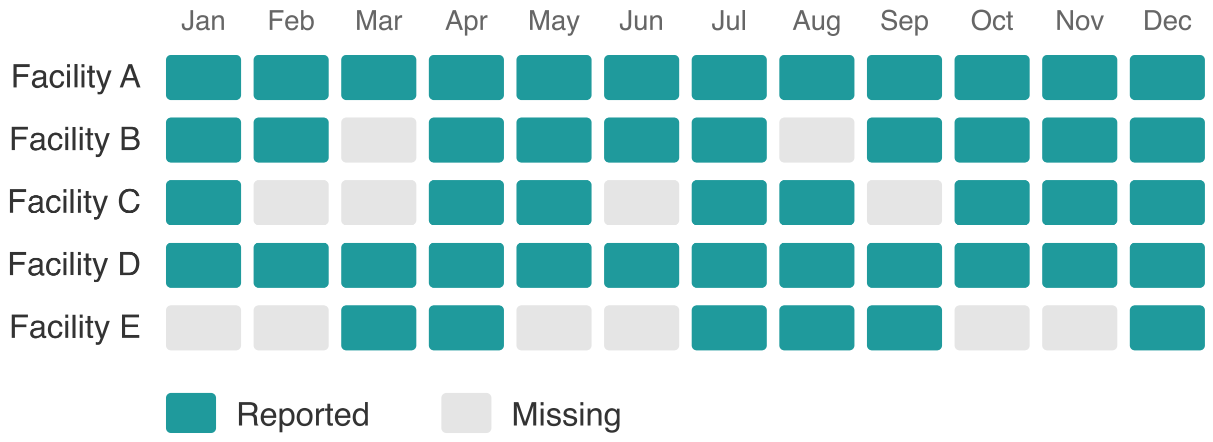
Why it matters:

- Higher completeness improves data reliability
- Stability over time strengthens trend analysis

Key distinction:
Indicator completeness ≠ reporting completeness. This metric examines specific data elements, not just whether the monthly form was submitted.

Region A: Indicator Completeness

5 health facilities reporting monthly on ANC1



5 facilities × 12 months = 60 expected
48 reported → 80% completeness

Definition of indicator completeness

For the FASTR analysis, completeness is defined as:

The percentage of reporting facilities each month out of the total number of facilities expected to report.

- A facility is deemed to be "reporting" if there is a non-missing, non-zero value recorded for the indicator and month
- A facility is expected to report if it has reported any volume for that indicator anytime within a year
- Facilities that do not report for six or more consecutive months at the beginning or end of their reporting period are classified as **inactive** rather than incomplete. This prevents penalizing facilities that have not yet begun reporting or have permanently ceased operations

Notes on completeness

- A high level of completeness does not necessarily indicate that the HMIS is representative of all service delivery in the country as some services may not be delivered in facilities, or some facilities may not report
- For countries where the DHIS2 system does not store 0's, indicator completeness may be underestimated if there are many low-volume facilities for a given indicator

Completeness: Percent of monthly values that are complete

For a given indicator in a given time period, the percent of monthly values that are complete:

% complete = # monthly values that are complete / total N of monthly values

Indicator Completeness

Percentage of facility-months with complete data, Jan 2022 to Apr 2025

	Antenatal care 1	Antenatal care 4	Institutional delivery	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG vaccine	Penta vaccine 1	Penta vaccine 3	Outpatient visit
District 001	93.9%	90.3%	83.3%	70.8%	82.8%	92.3%	91.9%	90.4%	94.9%
District 002	88.5%	85.9%	82.0%	64.2%	79.8%	90.8%	88.6%	88.1%	88.4%
District 003	89.8%	87.0%	85.4%	53.4%	78.5%	86.2%	86.3%	85.5%	88.3%
District 004	90.7%	82.6%	84.5%	67.6%	82.1%	89.1%	90.4%	89.7%	90.4%
District 005	89.7%	84.0%	81.3%	60.4%	80.9%	84.3%	84.2%	83.6%	89.5%
District 006	85.5%	79.4%	74.5%	56.8%	74.5%	85.3%	82.9%	79.2%	85.6%
District 007	97.2%	93.8%	91.8%	73.5%	90.8%	91.3%	95.2%	87.9%	97.0%
District 008	93.1%	87.6%	85.5%	44.6%	71.8%	79.9%	87.2%	87.5%	95.0%
District 009	95.7%	75.8%	85.0%	34.9%	77.6%	83.5%	92.2%	78.1%	95.9%
District 010	99.2%	99.6%	95.8%	98.4%	94.2%	94.6%	93.2%	88.7%	100.0%
District 011	98.0%	95.2%	94.4%	66.6%	91.7%	96.1%	95.4%	91.8%	98.2%
District 012	92.6%	86.4%	83.7%	64.5%	84.7%	84.7%	87.2%	86.7%	92.6%
District 013	93.4%	90.8%	90.7%	77.1%	88.2%	90.3%	90.8%	87.7%	93.1%
District 014	88.8%	79.1%	81.2%	77.9%	81.2%	90.0%	89.8%	86.3%	91.2%
District 015	93.0%	88.2%	85.3%	63.9%	83.1%	86.7%	88.8%	88.7%	93.1%
District 016	96.0%	85.1%	92.0%	65.7%	91.6%	96.1%	95.5%	88.6%	96.8%
District 017	90.1%	84.8%	90.4%	53.4%	83.0%	72.7%	79.3%	76.0%	89.5%
District 018	97.1%	95.9%	93.2%	79.8%	92.3%	95.4%	95.8%	92.9%	97.9%

90% or above
80% to 89%
Below 80%

Higher completeness improves the reliability of the data, especially when completeness is stable over time. Completeness is defined as the percentage of reporting facilities each month out of the total number of facilities expected to report. A facility is expected to report if it has reported any volume for each indicator anytime within a year. A high completeness does not indicate that the HMIS is representative of all service delivery in the country, as some services may not be delivered in facilities, or some facilities may not report.

Outliers

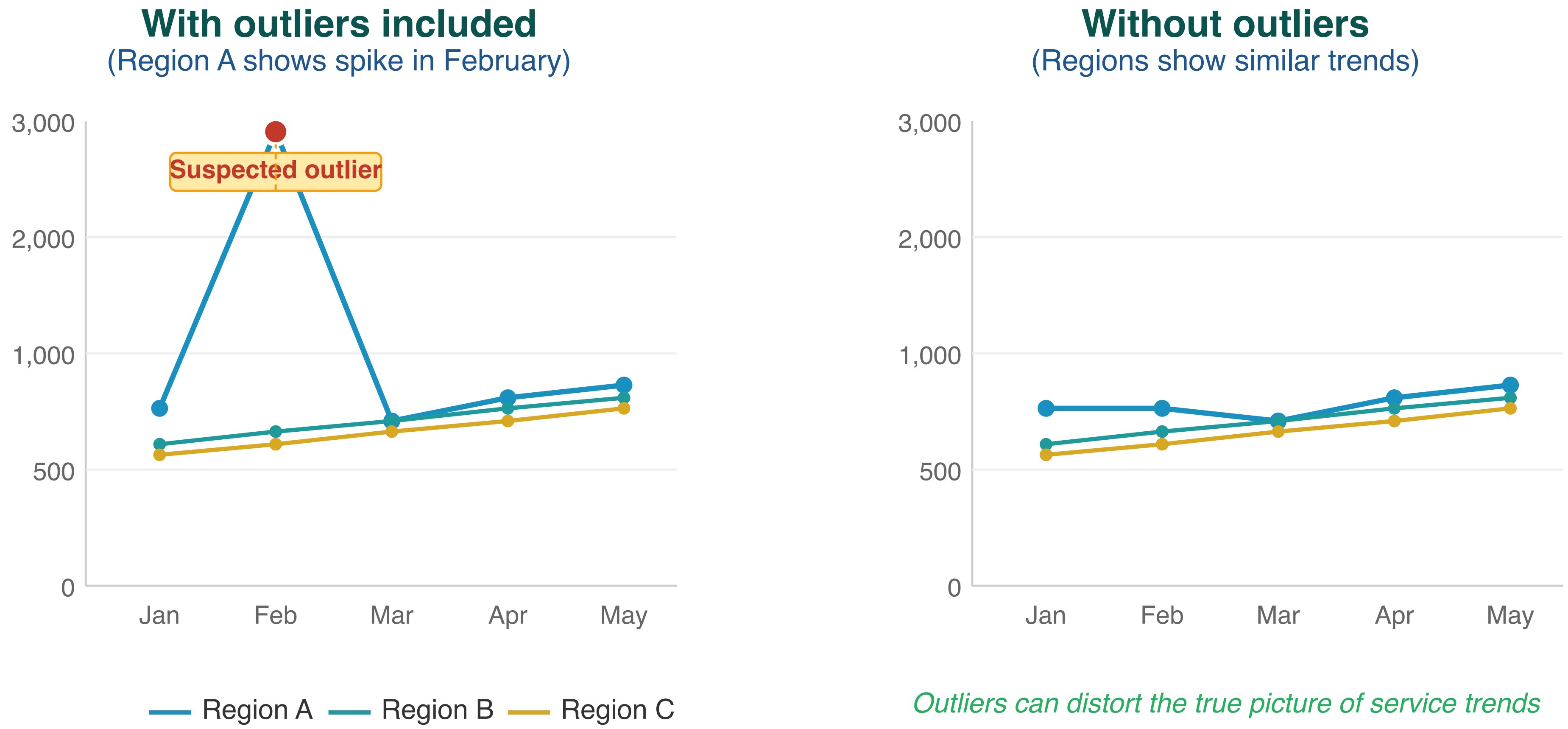
The presence of outliers examines whether a data point in a series of values is extreme (either abnormally high or low) in relation to others in the series.

Outliers can be the result of changes in programmatic activities (such as an intensified campaign) or can be data quality problems.

For the FASTR analysis, we identify outliers which are suspiciously high values compared to the usual volume of services reported by the facility (e.g., low values are not identified as outliers in the FASTR analysis).

Outlier illustration

Region A displays an anomalous spike in February that substantially exceeds values reported by other regions — indicative of a data entry error or reporting issue.



Outliers can distort the true picture of service trends

Outlier detection methodology

Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator.

An outlier is defined as:

A value greater than **10 times the median absolute deviation (MAD)** from the monthly median value for the indicator in each time period, **OR** a value for which the proportional contribution in volume for a facility, indicator, and time period is **greater than 80%**

AND for which:

- The volume is **greater than or equal to the median**
- The volume is **not missing**
- The volume is **greater than 100**

Outliers: Percent of monthly values that are outliers

For a given indicator in a given time period, the percent of monthly values that are outliers:

% outliers = # monthly values that are outliers / total N of monthly values

Outliers

Percentage of facility-months that are outliers, May 2024 to Apr 2025

	Antenatal care 1	Antenatal care 4	Institutional delivery	Postnatal care 1 (newborns)	Postnatal care 1 (mothers)	BCG vaccine	Penta vaccine 1	Penta vaccine 3	Outpatient visit
Region 001	0.6%	0.2%	0.2%	1.9%	1.2%	0.6%	0.9%	0.6%	0.9%
Region 002	0.6%	0.0%	1.2%	1.0%	1.8%	0.2%	0.2%	0.1%	2.7%
Region 003	0.5%	0.4%	0.1%	0.0%	0.1%	0.1%	0.4%	0.6%	1.4%
Region 004	0.4%	0.0%	0.0%	0.8%	0.0%	0.5%	0.8%	0.9%	0.1%
Region 005	0.5%	0.9%	0.5%	0.8%	0.5%	0.6%	0.8%	0.3%	3.3%
Region 006	0.4%	0.0%	0.1%	0.3%	0.1%	1.3%	2.3%	2.5%	0.6%

3% or above
1% to 2%
Below 1%

Outliers are reports which are suspiciously high compared to the usual volume reported by the facility in other months. Outliers are identified by assessing the within-facility variation in monthly reporting for each indicator. Outliers are defined observations which are greater than 10 times the median absolute deviation (MAD) from the monthly median value for the indicator in each time period, OR a value for which the proportional contribution in volume for a facility, indicator, and time period is greater than 80%. Outliers are only identified for indicators where the volume is greater than or equal to the median, the volume is not missing, and the average volume is greater than 100.

Consistency between related indicators

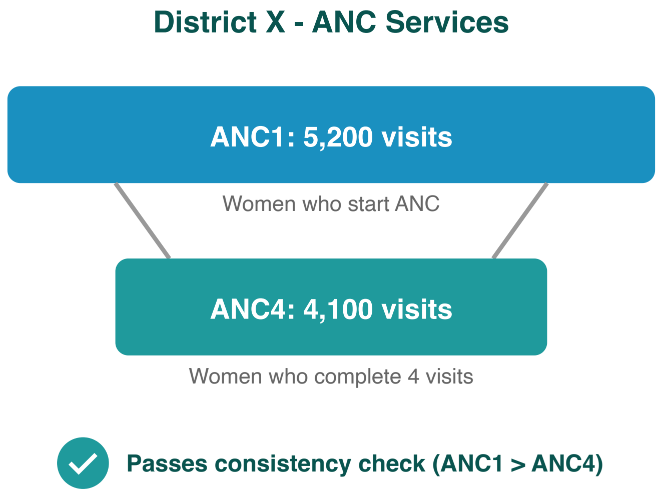
Program indicators with a predictable relationship are examined to determine whether the expected relationship exists between them. In other words, this process examines whether the observed relationship between the indicators, as shown in the reported data, is that which is expected.

Indicator pairs assessed

Indicator pair	Expected relationship
ANC1 / ANC4	Ratio should be ≥ 0.95
Penta1 / Penta3	Ratio should be ≥ 0.95
BCG / Facility delivery	Within 30% (≥0.7 and ≤1.3)

These pairs have expected relationships. We expect ANC1 > ANC4 since not all women complete four visits.

BCG is a birth dose vaccine so we expect similar numbers to facility deliveries, with a 30% tolerance for variability.

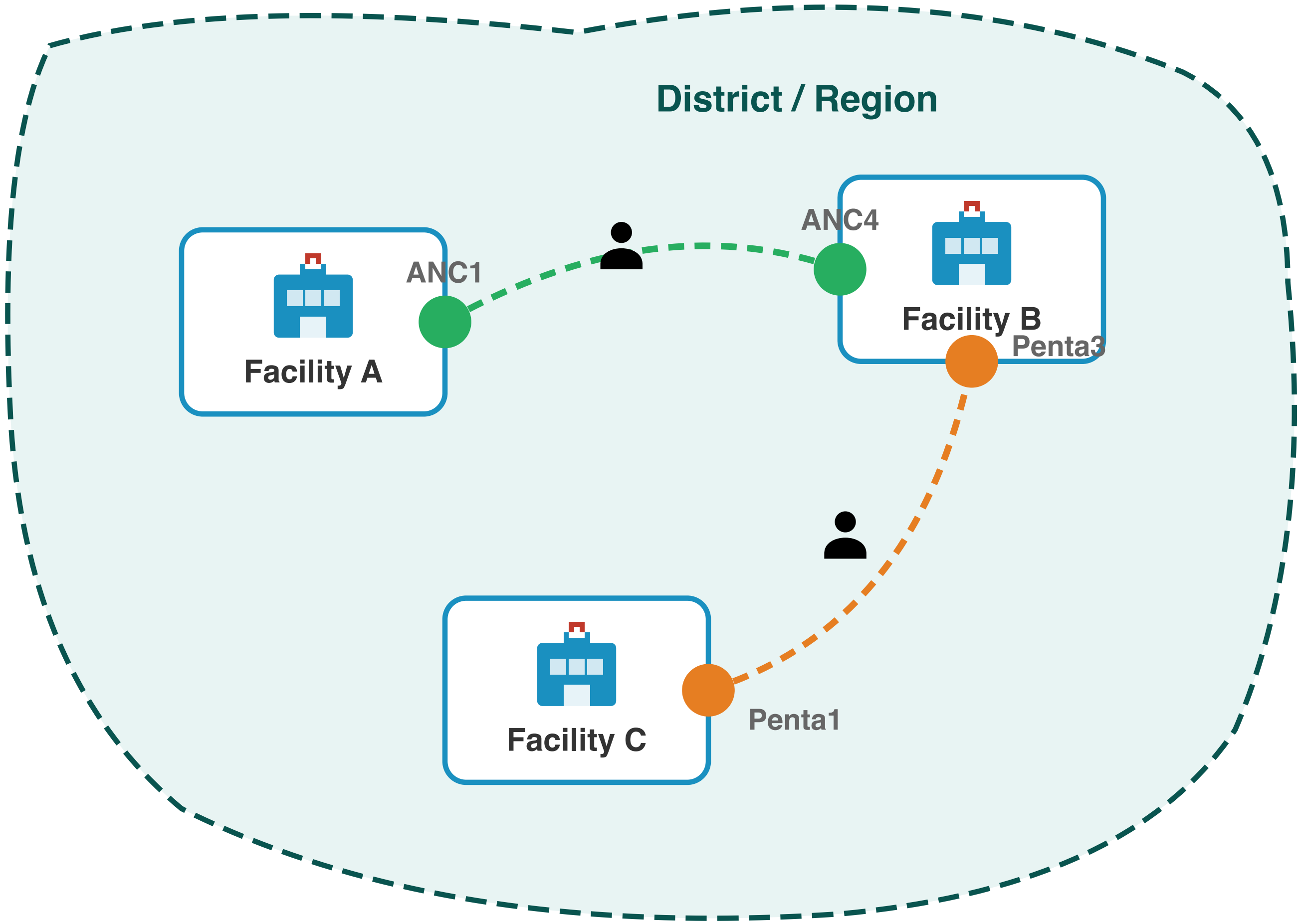


Why assess consistency at district level?

Patients often access different services at different facilities within a district:

- A woman may attend **ANC1** at a nearby health post, but travel to a health centre for **ANC4**
- A child may receive **Penta1** at a local clinic, but complete **Penta3** at a district hospital

Checking consistency at the facility level would miss these patterns. Aggregating to district level captures the complete picture of service utilization within a geographic area.



Internal consistency: FASTR output

Internal consistency

Percentage of sub-national areas meeting consistency benchmarks, May 2024 to Apr 2025

	ANC1 is larger than ANC4	Delivery is approximately equal to BCG	Penta1 is larger than Penta3
Region 001	100.0%	0.0%	97.2%
Region 002	100.0%	0.0%	100.0%
Region 003	100.0%	45.8%	83.3%
Region 004	100.0%	37.5%	87.5%
Region 005	100.0%	0.0%	87.5%
Region 006	100.0%	49.0%	84.4%

- 90% or above
- 80% to 89%
- Below 80%

Internal consistency assesses the plausibility of reported data based on related indicators. Consistency metrics are approximate - depending on timing and seasonality, indicator definitions, and the nature of service delivery and reporting, values may be expected to sit outside plausible ranges. Indicators which are similar are expected to have roughy the same volume over the year (within a 30% margin). The data in this analysis is adjusted for outliers.

Data quality summary score

A composite measure of data quality provides an overall view of how well a dataset meets quality standards.

By integrating multiple dimensions of data quality into a single score, it simplifies the interpretation of detailed information from several measures. This allows health systems to quickly assess the reliability of data, making it easier to identify trends and issues at a glance.

Definition of adequate data quality

For the FASTR analysis, we defined adequate data quality as:

- No missing indicator data for OPD, Penta1, and ANC1, where available, **AND**
- No outliers for OPD, Penta1, and ANC1, where available, **AND**
- Consistent reporting between Penta1/Penta3 and ANC1/ANC4

Overall DQA score: Percent of monthly values meeting all criteria

For a given indicator in a given time period, the percent of monthly values meeting all DQA criteria:

% adequate quality = # monthly values meeting all criteria / total N of monthly values

Overall DQA score

Percentage of facility-months with adequate data quality over time

	2022	2023	2024	2025
Region 001	60.8%	76.6%	76.4%	84.4%
Region 002	55.3%	61.2%	58.2%	52.0%
Region 003	69.3%	73.4%	60.6%	47.0%
Region 004	54.6%	70.7%	70.0%	84.9%
Region 005	57.9%	69.5%	56.6%	55.4%
Region 006	74.0%	84.7%	72.2%	71.7%

80% or above
70% to 79%
Below 70%

Adequate data quality is defined as: 1) No missing data or outliers for OPD, Penta1, and ANC1, where available 2) Consistent reporting between Penta1/Penta3 and ANC1/ANC4.

Mean DQA score: How close are we to adequate quality?

The mean DQA score shows how close a facility's data is to meeting all quality criteria. A score of **100% means the data passes** all DQA checks—no missing values, no outliers, and consistent reporting.

Mean DQA = (completeness & outlier score + consistency score) / 2

Mean DQA score

Average data quality score across facility-months

	2022	2023	2024	2025
Region 001	90.2%	95.5%	95.5%	96.5%
Region 002	87.3%	89.0%	88.2%	85.6%
Region 003	92.8%	94.8%	91.0%	84.3%
Region 004	88.0%	93.9%	93.0%	97.0%
Region 005	89.5%	93.6%	91.1%	88.6%
Region 006	93.2%	96.3%	93.4%	93.4%

80% or above
70% to 79%
Below 70%

Items included in the DQA score include: No missing data for 1) OPD, 2) Penta1, and 3) ANC1, where available; No outliers for 4) OPD, 5) Penta1, and 6) ANC1, where available; Consistent reporting between 7) Penta1/Penta3, 8) ANC1/ANC4, 9)BCG/Delivery, where available.

Contact Information

FASTR Team

Email:

Website: https://www.globalfinancingfacility.org/